

## Chapter 6.7

### *Provisions for the design, construction, inspection and testing of portable tanks and multiple-element gas containers (MEGCs)*

**Note 1:** The provisions of this chapter also apply to road tank vehicles to the extent indicated in chapter 6.8.

**Note 2:** The provisions of this chapter also apply to portable tanks with shells made of fibre-reinforced plastics (FRP) to the extent indicated in chapter 6.10.

#### 6.7.1 Application and general provisions

**6.7.1.1** The provisions of this chapter apply to portable tanks intended for the transport of dangerous goods, and to MEGCs intended for the transport of non-refrigerated gases of class 2, by all modes of transport. In addition to the provisions of this chapter, unless otherwise specified, the applicable provisions of the *International Convention for Safe Containers, 1972* (CSC Convention), as amended, shall be fulfilled by any multimodal portable tank or MEGC which meets the definition of a “container” within the terms of that Convention. Additional provisions may apply to offshore portable tanks that are handled in open seas.

**6.7.1.1.1** The *International Convention for Safe Containers, 1972* (CSC Convention) does not apply to offshore tank-containers that are handled in open seas. The design and testing of offshore tank-containers shall take into account the dynamic lifting and impact forces that may occur when a tank is handled in open seas in adverse weather and sea conditions. The provisions for such tanks shall be determined by the approving competent authority (see also *Guidelines for the approval of offshore containers handled in open seas* (MSC/Circ.860)).

**6.7.1.2** In recognition of scientific and technological advances, the technical provisions of this chapter may be varied by alternative arrangements. These alternative arrangements shall offer a level of safety not less than that given by the provisions of this chapter with respect to the compatibility with substances transported and the ability of the portable tank to withstand impact, loading and fire conditions. For international transport, alternative arrangement portable tanks or MEGCs shall be approved by the applicable competent authorities.

**6.7.1.3** When a substance is not assigned a portable tank instruction (T1 to T75) in the Dangerous Goods List in chapter 3.2, interim approval for transport may be issued by the competent authority of the country of origin. The approval shall be included in the documentation of the consignment and contain, as a minimum, the information normally provided in the portable tank instructions and the conditions under which the substance shall be transported. Appropriate measures shall be initiated by the competent authority to include the assignment in the Dangerous Goods List.

#### 6.7.2 Provisions for the design, construction, inspection and testing of portable tanks intended for the transport of substances of class 1 and classes 3 to 9

##### 6.7.2.1 Definitions

For the purposes of this section:

*Design pressure* means the pressure to be used in calculations required by a recognized pressure-vessel code. The design pressure shall be not less than the highest of the following pressures:

- .1 the maximum effective gauge pressure allowed in the shell during filling or discharge; or
- .2 the sum of:
  - .1 the absolute vapour pressure (in bar) of the substance at 65°C (or at the highest temperature during filling, discharge or transport for substances which are filled, discharged or transported over 65°C), minus 1 bar;
  - .2 the partial pressure (in bar) of air or other gases in the ullage space, being determined by a maximum ullage temperature of 65°C and a liquid expansion due to an increase in mean bulk temperature of  $t_r - t_f$  ( $t_f$  = filling temperature, usually 15°C;  $t_r$  = 50°C, maximum mean bulk temperature); and

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- .3 a head pressure determined on the basis of the static forces specified in 6.7.2.2.12, but not less than 0.35 bar; or
- .3 two thirds of the minimum test pressure specified in the applicable portable tank instruction in 4.2.5.2.6;

*Design temperature range* for the shell shall be  $-40^{\circ}\text{C}$  to  $50^{\circ}\text{C}$  for substances transported under ambient conditions. For the other substances filled, discharged or transported above  $50^{\circ}\text{C}$ , the design temperature shall not be less than the maximum temperature of the substance during filling, discharge or transport. More severe design temperatures shall be considered for portable tanks subjected to severe climatic conditions;

*Fine grain steel* means steel which has a ferritic grain size of 6 or finer when determined in accordance with ASTM E 112-96 or as defined in EN 10028-3, Part 3;

*Fusible element* means a non-reclosable pressure relief device that is thermally actuated;

*Leakproofness test* means a test using gas, subjecting the shell and its service equipment to an effective internal pressure of not less than 25% of the MAWP;

*Maximum allowable working pressure* (MAWP) means a pressure that shall be not less than the highest of the following pressures measured at the top of the shell while in operating position:

- .1 the maximum effective gauge pressure allowed in the shell during filling or discharge; or
- .2 the maximum effective gauge pressure to which the shell is designed, which shall be not less than the sum of:
  - .1 the absolute vapour pressure (in bar) of the substance at  $65^{\circ}\text{C}$  (or at the highest temperature during filling, discharge or transport for substances which are filled, discharged or transported over  $65^{\circ}\text{C}$ ) minus 1 bar; and
  - .2 the partial pressure (in bar) of air or other gases in the ullage space, being determined by a maximum ullage temperature of  $65^{\circ}\text{C}$  and a liquid expansion due to an increase in mean bulk temperature of  $t_r - t_f$  ( $t_f$  = filling temperature, usually  $15^{\circ}\text{C}$ ;  $t_r$  =  $50^{\circ}\text{C}$ , maximum mean bulk temperature);

*Maximum permissible gross mass* (MPGM) means the sum of the tare mass of the portable tank and the heaviest load authorized for transport;

*Mild steel* means a steel with a guaranteed minimum tensile strength of  $360\text{ N/mm}^2$  to  $440\text{ N/mm}^2$  and a guaranteed minimum elongation at fracture conforming to 6.7.2.3.3.3;

*Offshore portable tank* means a portable tank specially designed for repeated use for transport of dangerous goods to, from and between offshore facilities. An offshore portable tank is designed and constructed in accordance with the *Guidelines for the approval of offshore containers handled in open seas* (MSC/Circ.860).

*Portable tank* means a multimodal tank used for the transport of substances of class 1 and classes 3 to 9. The portable tank includes a shell fitted with service equipment and structural equipment necessary for the transport of dangerous substances. The portable tank shall be capable of being filled and discharged without the removal of its structural equipment. It shall possess stabilizing members external to the shell, and shall be capable of being lifted when full. It shall be designed primarily to be loaded onto a vehicle or ship and shall be equipped with skids, mountings or accessories to facilitate mechanical handling. Road tank-vehicles, rail tank-wagons, non-metallic tanks (except FRP portable tanks, see chapter 6.10), gas cylinders, large receptacles, and intermediate bulk containers (IBCs) are not considered to fall within this definition;

*Reference steel* means a steel with a tensile strength of  $370\text{ N/mm}^2$  and an elongation at fracture of 27%;

*Service equipment* means measuring instruments and filling, discharge, venting, safety, heating, cooling and insulating devices;

*Shell* means the part of the portable tank which retains the substance intended for transport (tank proper), including openings and their closures, but does not include service equipment or external structural equipment;

*Structural equipment* means the reinforcing, fastening, protective and stabilizing members external to the shell;

*Test pressure* means the maximum gauge pressure at the top of the shell during the hydraulic pressure test, equal to not less than 1.5 times the design pressure. The minimum test pressure for portable tanks intended for specific substances is specified in the applicable portable tank instruction in 4.2.5.2.6.

### 6.7.2.2 General design and construction provisions

- 6.7.2.2.1 Shells shall be designed and constructed in accordance with the provisions of a pressure-vessel code recognized by the competent authority. Shells shall be made of metallic materials suitable for forming. The materials shall, in principle, conform to national or international material standards. For welded shells, only a material whose weldability has been fully demonstrated shall be used. Welds shall be skillfully made and

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afford complete safety. When the manufacturing process or the materials make it necessary, the shells shall be suitably heat-treated to guarantee adequate toughness in the weld and in the heat-affected zones. In choosing the material, the design temperature range shall be taken into account with respect to risk of brittle fracture, to stress corrosion cracking and to resistance to impact. When fine grain-steel is used, the guaranteed value of the yield strength shall be not more than 460 N/mm<sup>2</sup> and the guaranteed value of the upper limit of the tensile strength shall be not more than 725 N/mm<sup>2</sup> according to the material specification. Aluminium may only be used as a construction material when indicated in a portable tank special provision assigned to a specific substance in the Dangerous Goods List or when approved by the competent authority. When aluminium is authorized, it shall be insulated to prevent significant loss of physical properties when subjected to a heat load of 110 kW/m<sup>2</sup> for a period of not less than 30 minutes. The insulation shall remain effective at all temperatures less than 649°C and shall be jacketed with a material with a melting point of not less than 700°C. Portable tank materials shall be suitable for the external environment in which they may be transported.

- 6.7.2.2.2** Portable tank shells, fittings, and pipework shall be constructed from materials which are:
- .1 substantially immune to attack by the substance(s) intended to be transported; or
  - .2 properly passivated or neutralized by chemical reaction; or
  - .3 lined with corrosion-resistant material directly bonded to the shell or attached by equivalent means.
- 6.7.2.2.3** Gaskets shall be made of materials not subject to attack by the substance(s) intended to be transported.
- 6.7.2.2.4** When shells are lined, the lining shall be substantially immune to attack by the substance(s) intended to be transported, homogeneous, non-porous, free from perforations, sufficiently elastic and compatible with the thermal expansion characteristics of the shell. The lining of every shell, shell fittings and piping shall be continuous, and shall extend around the face of any flange. Where external fittings are welded to the tank, the lining shall be continuous through the fitting and around the face of external flanges.
- 6.7.2.2.5** Joints and seams in the lining shall be made by fusing the material together or by other equally effective means.
- 6.7.2.2.6** Contact between dissimilar metals which could result in damage by galvanic action shall be avoided.
- 6.7.2.2.7** The materials of the portable tank, including any devices, gaskets, linings and accessories, shall not adversely affect the substance(s) intended to be transported in the portable tank.
- 6.7.2.2.8** Portable tanks shall be designed and constructed with supports to provide a secure base during transport and with suitable lifting and tie-down attachments.
- 6.7.2.2.9** Portable tanks shall be designed to withstand, without loss of contents, at least the internal pressure due to the contents and the static, dynamic and thermal loads during normal conditions of handling and transport. The design shall demonstrate that the effects of fatigue, caused by repeated application of these loads through the expected life of the portable tank, have been taken into account.
- 6.7.2.2.9.1** For portable tanks that are intended for use as offshore tank-containers, the dynamic stresses imposed by handling in open seas shall be taken into account.
- 6.7.2.2.10** A shell which is to be equipped with a vacuum-relief device shall be designed to withstand, without permanent deformation, an external pressure of not less than 0.21 bar above the internal pressure. The vacuum-relief device shall be set to relieve at a vacuum setting not greater than –0.21 bar unless the shell is designed for a higher external overpressure, in which case the vacuum-relief pressure of the device to be fitted shall be not greater than the tank design vacuum pressure. A shell used for the transport of solid substances of packing groups II or III only which do not liquefy during transport may be designed for a lower external pressure, subject to competent authority's approval. In this case, the vacuum-relief device shall be set to relieve at this lower pressure. A shell that is not to be fitted with a vacuum-relief device shall be designed to withstand, without permanent deformation, an external pressure of not less than 0.4 bar above the internal pressure.
- 6.7.2.2.11** Vacuum-relief devices used on portable tanks intended for the transport of substances meeting the flashpoint criteria of class 3, including elevated-temperature substances transported at or above their flashpoint, shall prevent the immediate passage of flame into the shell, or the portable tank shall have a shell capable of withstanding, without leakage, an internal explosion resulting from the passage of flame into the shell.
- 6.7.2.2.12** Portable tanks and their fastenings shall, under the maximum permissible load, be capable of absorbing the following separately applied static forces:
- .1 in the direction of travel: twice the MPGM multiplied by the acceleration due to gravity (g);\*
  - .2 horizontally at right angles to the direction of travel: the MPGM (when the direction of travel is not clearly determined, the forces shall be equal to twice the MPGM) multiplied by the acceleration due to gravity (g);\*

\* For calculation purposes,  $g = 9.81 \text{ m/s}^2$ .

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- .3 vertically upwards: the MPGM multiplied by the acceleration due to gravity (g);\* and
  - .4 vertically downwards: twice the MPGM (total loading including the effect of gravity) multiplied by the acceleration due to gravity (g).\*
- 6.7.2.2.13 Under each of the forces in 6.7.2.2.12, the safety factor to be observed shall be as follows:
- .1 for metals having a clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed yield strength; or
  - .2 for metals with no clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed 0.2% proof strength and, for austenitic steels, the 1% proof strength.
- 6.7.2.2.14 The value of yield strength or proof strength shall be the value according to national or international material standards. When austenitic steels are used, the specified minimum values of yield strength or proof strength according to the material standards may be increased by up to 15% when these greater values are attested in the material inspection certificate. When no material standard exists for the metal in question, the value of yield strength or proof strength used shall be approved by the competent authority.
- 6.7.2.2.15 Portable tanks shall be capable of being electrically earthed when intended for the transport of substances meeting the flashpoint criteria of class 3, including elevated-temperature substances transported above their flashpoint. Measures shall be taken to prevent dangerous electrostatic discharge.
- 6.7.2.2.16 When required for certain substances by the applicable portable tank instruction indicated in column 13 of the Dangerous Goods List, or by a portable tank special provision indicated in column 14 of the Dangerous Goods List, portable tanks shall be provided with additional protection, which may take the form of additional shell thickness or a higher test pressure, the additional shell thickness or higher test pressure being determined in the light of the inherent risks associated with the transport of the substances concerned.
- 6.7.2.2.17 Thermal insulation directly in contact with the shell intended for substances transported at elevated temperature shall have an ignition temperature at least 50°C higher than the maximum design temperature of the tank.
- 6.7.2.3 **Design criteria**
- 6.7.2.3.1 Shells shall be of a design capable of being stress-analysed mathematically or experimentally by resistance strain gauges, or by other methods approved by the competent authority.
- 6.7.2.3.2 Shells shall be designed and constructed to withstand a hydraulic test pressure not less than 1.5 times the design pressure. Specific provisions are laid down for certain substances in the applicable portable tank instruction indicated in column 13 of the Dangerous Goods List and described in 4.2.5.2.6 or by a portable tank special provision indicated in column 14 of the Dangerous Goods List and described in 4.2.5.3. The minimum shell thickness shall not be less than that specified for these tanks in 6.7.2.4.1 to 6.7.2.4.10.
- 6.7.2.3.3 For metals exhibiting a clearly defined yield point or characterized by a guaranteed proof strength (0.2% proof strength, generally, or 1% proof strength for austenitic steels), the primary membrane stress  $\sigma$  (sigma) in the shell shall not exceed  $0.75R_e$  or  $0.50R_m$ , whichever is lower, at the test pressure, where:
- $R_e$  = yield strength in N/mm<sup>2</sup>, or 0.2% proof strength or, for austenitic steels, 1% proof strength;
  - $R_m$  = minimum tensile strength in N/mm<sup>2</sup>.
- 6.7.2.3.3.1 The values of  $R_e$  and  $R_m$  to be used shall be the specified minimum values according to national or international material standards. When austenitic steels are used, the specified minimum values for  $R_e$  and  $R_m$  according to the material standards may be increased by up to 15% when these greater values are attested in the material inspection certificate. When no material standard exists for the metal in question, the values of  $R_e$  and  $R_m$  used shall be approved by the competent authority or its authorized body.
- 6.7.2.3.3.2 Steels which have a  $R_e/R_m$  ratio of more than 0.85 are not allowed for the construction of welded shells. The values of  $R_e$  and  $R_m$  to be used in determining this ratio shall be the values specified in the material inspection certificate.
- 6.7.2.3.3.3 Steels used in the construction of shells shall have an elongation at fracture, in %, of not less than  $10,000/R_m$  with an absolute minimum of 16% for fine-grain steels and 20% for other steels. Aluminium and aluminium alloys used in the construction of shells shall have an elongation at fracture, in %, of not less than  $10,000/6R_m$  with an absolute minimum of 12%.
- 6.7.2.3.3.4 For the purpose of determining actual values for materials, it shall be noted that for sheet metal, the axis of the tensile test specimen shall be at right angles (transversely) to the direction of rolling. The permanent elongation at fracture shall be measured on test specimens of rectangular cross-section in accordance with ISO 6892:1998 using a 50 mm gauge length.

\* For calculation purposes,  $g = 9.81 \text{ m/s}^2$ .

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**6.7.2.4.1** The minimum shell thickness shall be the greater thickness based on:

- .1 the minimum thickness determined in accordance with the provisions of 6.7.2.4.2 to 6.7.2.4.10;
- .2 the minimum thickness determined in accordance with the recognized pressure-vessel code, including the provisions in 6.7.2.3; and
- .3 the minimum thickness specified in the applicable portable tank instruction indicated in column 13 of the Dangerous Goods List, or by a portable tank special provision indicated in column 14.

**6.7.2.4.2** The cylindrical portions, ends (heads) and manhole covers of shells not more than 1.80 m in diameter shall be not less than 5 mm thick in the reference steel or of equivalent thickness in the metal to be used. Shells more than 1.80 m in diameter shall be not less than 6 mm thick in the reference steel or of equivalent thickness in the metal to be used, except that for powdered or granular solid substances of packing group II or III the minimum thickness requirement may be reduced to not less than 5 mm thick in the reference steel or of equivalent thickness in the metal to be used.

**6.7.2.4.3** When additional protection against shell damage is provided, portable tanks with test pressures less than 2.65 bar may have the minimum shell thickness reduced, in proportion to the protection provided, as approved by the competent authority. However, shells not more than 1.80 m in diameter shall be not less than 3 mm thick in the reference steel or of equivalent thickness in the metal to be used. Shells more than 1.80 m in diameter shall be not less than 4 mm thick in the reference steel or of equivalent thickness in the metal to be used.

**6.7.2.4.4** The cylindrical portions, ends (heads) and manhole covers of all shells shall be not less than 3 mm thick regardless of the material of construction.

**6.7.2.4.5** The additional protection referred to in 6.7.2.4.3 may be provided by overall external structural protection, such as suitable “sandwich” construction with the outer sheathing (jacket) secured to the shell, double-wall construction or by enclosing the shell in a complete framework with longitudinal and transverse structural members.

**6.7.2.4.6** The equivalent thickness of a metal other than the thickness prescribed for the reference steel in 6.7.2.4.3 shall be determined using the following equation:

$$e_1 = \frac{21.4 \times e_0}{\sqrt[3]{R_{m1} \times A_1}}$$

where:

- $e_1$  = required equivalent thickness (in mm) of the metal to be used;
- $e_0$  = minimum thickness (in mm) of the reference steel specified in the applicable portable tank instruction or by a portable tank special provision indicated in column 13 or 14 of the Dangerous Goods List;
- $R_{m1}$  = guaranteed minimum tensile strength (in N/mm<sup>2</sup>) of the metal to be used (see 6.7.2.3.3);
- $A_1$  = guaranteed minimum elongation at fracture (in %) of the metal to be used according to national or international standards.

**6.7.2.4.7** When, in the applicable portable tank instruction in 4.2.5.2.6, a minimum thickness of 8 mm, 10 mm or 12 mm is specified, it shall be noted that these thicknesses are based on the properties of the reference steel and a shell diameter of 1.80 m. When a metal other than mild steel (see 6.7.2.1) is used or the shell has a diameter of more than 1.80 m, the thickness shall be determined using the following equation:

$$e_1 = \frac{21.4 \times e_0 d_1}{1.8 \sqrt[3]{R_{m1} \times A_1}}$$

where:

- $e_1$  = required equivalent thickness (in mm) of the metal to be used;
- $e_0$  = minimum thickness (in mm) of the reference steel specified in the applicable portable tank instruction or by a portable tank special provision indicated in column 13 or 14 of the Dangerous Goods List;
- $d_1$  = diameter of the shell (in m), but not less than 1.80 m;
- $R_{m1}$  = guaranteed minimum tensile strength (in N/mm<sup>2</sup>) of the metal to be used (see 6.7.2.3.3);
- $A_1$  = guaranteed minimum elongation at fracture (in %) of the metal to be used according to national or international standards.

**6.7.2.4.8** In no case shall the wall thickness be less than that prescribed in 6.7.2.4.2, 6.7.2.4.3 and 6.7.2.4.4. All parts of the shell shall have a minimum thickness as determined by 6.7.2.4.2 to 6.7.2.4.4. This thickness shall be exclusive of any corrosion allowance.

**6.7.2.4.9** When mild steel is used (see 6.7.2.1), calculation using the equation in 6.7.2.4.6 is not required.

**6.7.2.4.10** There shall be no sudden change of plate thickness at the attachment of the ends (heads) to the cylindrical portion of the shell.



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**6.7.2.5 Service equipment**

- 6.7.2.5.1** Service equipment shall be so arranged as to be protected against the risk of being wrenched off or damaged during handling and transport. When the connection between the frame and the shell allows relative movement between the sub-assemblies, the equipment shall be so fastened as to permit such movement without risk of damage to working parts. The external discharge fittings (pipe sockets, shut-off devices), the internal stop-valve and its seating shall be protected against the danger of being wrenched off by external forces (for example, by using shear sections). The filling and discharge devices (including flanges or threaded plugs) and any protective caps shall be capable of being secured against unintended opening.
- 6.7.2.5.1.1** For offshore tank-containers, where positioning of service equipment and the design and strength of protection for such equipment is concerned, the increased danger of impact damage when handling such tanks in open seas shall be taken into account.
- 6.7.2.5.2** All openings in the shell, intended for filling or discharging the portable tank, shall be fitted with a manually operated stop-valve located as close to the shell as reasonably practicable. Other openings, except for openings leading to venting or pressure relief devices, shall be equipped with either a stop-valve or another suitable means of closure located as close to the shell as reasonably practicable.
- 6.7.2.5.3** All portable tanks shall be fitted with a manhole or other inspection openings of a suitable size to allow for internal inspection and adequate access for maintenance and repair of the interior. Compartmented portable tanks shall have a manhole or other inspection openings for each compartment.
- 6.7.2.5.4** As far as reasonably practicable, external fittings shall be grouped together. For insulated portable tanks, top fittings shall be surrounded by a spill-collection reservoir with suitable drains.
- 6.7.2.5.5** Each connection to a portable tank shall be clearly marked to indicate its function.
- 6.7.2.5.6** Each stop-valve or other means of closure shall be designed and constructed to a rated pressure not less than the MAWP of the shell, taking into account the temperatures expected during transport. All stop-valves with screwed spindles shall close by a clockwise motion of the handwheel. For other stop-valves, the position (open and closed) and direction of closure shall be clearly indicated. All stop-valves shall be designed to prevent unintentional opening.
- 6.7.2.5.7** No moving parts, such as covers, components of closures, etc., shall be made of unprotected corrodible steel when they are liable to come into frictional or percussive contact with aluminium portable tanks intended for the transport of substances meeting the flashpoint criteria of class 3, including elevated-temperature substances transported above their flashpoint.
- 6.7.2.5.8** Piping shall be designed, constructed and installed so as to avoid the risk of damage due to thermal expansion and contraction, mechanical shock and vibration. All piping shall be of a suitable metallic material. Welded pipe joints shall be used wherever possible.
- 6.7.2.5.9** Joints in copper tubing shall be brazed or have an equally strong metal union. The melting point of brazing materials shall be no lower than 525°C. The joints shall not decrease the strength of the tubing, as may happen when cutting threads.
- 6.7.2.5.10** The burst pressure of all piping and pipe fittings shall be not less than the highest of four times the MAWP of the shell or four times the pressure to which it may be subjected in service by the action of a pump or other device (except pressure relief devices).
- 6.7.2.5.11** Ductile metals shall be used in the construction of valves and accessories.
- 6.7.2.5.12** The heating system shall be designed or controlled so that a substance cannot reach a temperature at which the pressure in the tank exceeds its MAWP or causes other hazards (e.g. dangerous thermal decomposition).
- 6.7.2.5.13** The heating system shall be designed or controlled so that power for internal heating elements is not available unless the heating elements are completely submerged. The temperature at the surface of the heating elements for internal heating equipment or the temperature at the shell for external heating equipment shall, in no case, exceed 80% of the auto-ignition temperature (in °C) of the substances carried.
- 6.7.2.5.14** If an electrical heating system is installed inside the tank, it shall be equipped with an earth leakage circuit breaker with a releasing current of less than 100 mA.
- 6.7.2.5.15** Electrical switch cabinets mounted to tanks shall not have a direct connection to the tank interior and shall provide protection of at least the equivalent of IP 56 according to IEC 144 or IEC 529.

**6.7.2.6 Bottom openings**

- 6.7.2.6.1** Certain substances shall not be transported in portable tanks with bottom openings. When the applicable portable tank instruction identified in the Dangerous Goods List and described in 4.2.5.2.6 indicates that bottom openings are prohibited, there shall be no openings below the liquid level of the shell when it is filled

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to its maximum permissible filling limit. When an existing opening is closed, it shall be accomplished by internally and externally welding one plate to the shell.

**6.7.2.6.2** Bottom discharge outlets for portable tanks carrying certain solid, crystallizable or highly viscous substances shall be equipped with not less than two serially fitted and mutually independent shut-off devices. The design of the equipment shall be to the satisfaction of the competent authority or its authorized body and shall include:

- .1 an external stop-valve, fitted as close to the shell as reasonably practicable, and so designed as to prevent any unintended opening through impact or other inadvertent act; and
- .2 a liquid-tight closure at the end of the discharge pipe, which may be a bolted blank flange or a screw cap.

**6.7.2.6.3** Every bottom discharge outlet, except as provided in 6.7.2.6.2, shall be equipped with three serially fitted and mutually independent shut-off devices. The design of the equipment shall be to the satisfaction of the competent authority or its authorized body and include:

- .1 a self-closing internal stop-valve, that is a stop-valve within the shell or within a welded flange or its companion flange, such that:
  - .1 the control devices for the operation of the valve are designed so as to prevent any unintended opening through impact or other inadvertent act;
  - .2 the valve may be operable from above or below;
  - .3 if possible, the setting of the valve (open or closed) shall be capable of being verified from the ground;
  - .4 except for portable tanks having a capacity of not more than 1000 L, it shall be possible to close the valve from an accessible position of the portable tank that is remote from the valve itself; and
  - .5 the valve shall continue to be effective in the event of damage to the external device for controlling the operation of the valve;
- .2 an external stop-valve fitted as close to the shell as reasonably practicable; and
- .3 a liquid-tight closure at the end of the discharge pipe, which may be a bolted blank flange or a screw cap.

**6.7.2.6.4** For a lined shell, the internal stop-valve required by 6.7.2.6.3.1 may be replaced by an additional external stop-valve. The manufacturer shall satisfy the provisions of the competent authority or its authorized body.

### **6.7.2.7 Safety relief devices**

**6.7.2.7.1** All portable tanks shall be fitted with at least one pressure relief device. All relief devices shall be designed, constructed and marked to the satisfaction of the competent authority or its authorized body.

### **6.7.2.8 Pressure relief devices**

**6.7.2.8.1** Every portable tank with a capacity not less than 1900 L and every independent compartment of a portable tank with a similar capacity shall be provided with one or more pressure relief devices of the spring-loaded type and may in addition have a frangible disc or fusible element in parallel with the spring-loaded devices except when prohibited by reference to 6.7.2.8.3 in the applicable portable tank instruction in 4.2.5.2.6. The pressure relief devices shall have sufficient capacity to prevent rupture of the shell due to over-pressurization or vacuum resulting from filling, from discharging, or from heating of the contents.

**6.7.2.8.2** Pressure relief devices shall be designed to prevent the entry of foreign matter, the leakage of liquid and the development of any dangerous excess pressure.

**6.7.2.8.3** When required for certain substances by the applicable portable tank instruction identified in the Dangerous Goods List and described in 4.2.5.2.6, portable tanks shall have a pressure relief device approved by the competent authority. Unless a portable tank in dedicated service is fitted with an approved relief device constructed of materials compatible with the load, the relief device shall comprise a frangible disc preceding a spring-loaded pressure relief device. When a frangible disc is inserted in series with the required pressure relief device, the space between the frangible disc and the pressure relief device shall be provided with a pressure gauge or suitable tell-tale indicator for the detection of disc rupture, pinholing, or leakage which could cause a malfunction of the pressure relief system. The frangible disc shall rupture at a nominal pressure 10% above the start-to-discharge pressure of the relief device.

**6.7.2.8.4** Every portable tank with a capacity less than 1900 L shall be fitted with a pressure relief device, which may be a frangible disc when this disc complies with the provisions of 6.7.2.11.1. When no spring-loaded pressure relief device is used, the frangible disc shall be set to rupture at a nominal pressure equal to the test pressure. In addition, fusible elements conforming to 6.7.2.10.1 may also be used.

**6.7.2.8.5** When the shell is fitted for pressure discharge, the inlet line shall be provided with a suitable pressure relief device set to operate at a pressure not higher than the MAWP of the shell, and a stop-valve shall be fitted as close to the shell as reasonably practicable.

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**6.7.2.9 Setting of pressure relief devices**

**6.7.2.9.1** It shall be noted that the pressure relief devices shall operate only in conditions of excessive rise in temperature, since the shell shall not be subject to undue fluctuations of pressure during normal conditions of transport (see 6.7.2.12.2).

**6.7.2.9.2** The required pressure relief device shall be set to start to discharge at a nominal pressure of five sixths of the test pressure for shells having a test pressure of not more than 4.5 bar and 110% of two thirds of the test pressure for shells having a test pressure of more than 4.5 bar. After discharge, the device shall close at a pressure not more than 10% below the pressure at which the discharge starts. The device shall remain closed at all lower pressures. This requirement does not prevent the use of vacuum relief or combination pressure relief and vacuum relief devices.

**6.7.2.10 Fusible elements**

**6.7.2.10.1** Fusible elements shall operate at a temperature between 100°C and 149°C on condition that the pressure in the shell at the fusing temperature will be not more than the test pressure. They shall be placed at the top of the shell with their inlets in the vapour space, and when used for transport safety purposes, they shall not be shielded from external heat. Fusible elements shall not be used on portable tanks with a test pressure which exceeds 2.65 bar unless specified by special provision TP36 in column 14 of the Dangerous Goods List of chapter 3.2. Fusible elements used on portable tanks intended for the transport of elevated-temperature substances shall be designed to operate at a temperature higher than the maximum temperature that will be experienced during transport and shall be to the satisfaction of the competent authority or its authorized body.

**6.7.2.11 Frangible discs**

**6.7.2.11.1** Except as specified in 6.7.2.8.3, frangible discs shall be set to rupture at a nominal pressure equal to the test pressure throughout the design temperature range. Particular attention shall be given to the provisions of 6.7.2.5.1 and 6.7.2.8.3 if frangible discs are used.

**6.7.2.11.2** Frangible discs shall be appropriate for the vacuum pressures which may be produced in the portable tank.

**6.7.2.12 Capacity of pressure relief devices**

**6.7.2.12.1** The spring-loaded pressure relief device required by 6.7.2.8.1 shall have a minimum cross-sectional flow area equivalent to an orifice of 31.75 mm diameter. Vacuum relief devices, when used, shall have a cross-sectional flow area not less than 284 mm<sup>2</sup>.

**6.7.2.12.2** The combined delivery capacity of the pressure relief system (taking into account the reduction of the flow when the portable tank is fitted with frangible discs preceding spring-loaded pressure relief devices or when the spring-loaded pressure relief devices are provided with a device to prevent the passage of the flame), in conditions of complete fire engulfment of the portable tank shall be sufficient to limit the pressure in the shell to 20% above the start-to-discharge pressure of the pressure-limiting device. Emergency pressure relief devices may be used to achieve the full relief capacity prescribed. These devices may be fusible, spring-loaded or frangible disc components, or a combination of spring-loaded and frangible disc devices. The total required capacity of the relief devices may be determined using the formula in 6.7.2.12.2.1 or the table in 6.7.2.12.2.3.

**6.7.2.12.2.1** To determine the total required capacity of the relief devices, which shall be regarded as being the sum of the individual capacities of all the contributing devices, the following formula shall be used:

$$Q = 12.4 \frac{FA^{0.82}}{LC} \sqrt{\frac{ZT}{M}}$$

where:

$Q$  = minimum required rate of discharge in cubic metres of air per second (m<sup>3</sup>/s) at standard conditions: 1 bar and 0°C (273 K);

$F$  = a coefficient with the following value:

for uninsulated shells,  $F = 1$

for insulated shells,  $F = U(649 - t)/13.6$  but in no case is less than 0.25

where:

$\Delta U$  = heat transfer coefficient of the insulation, in kW·m<sup>-2</sup>·K<sup>-1</sup>, at 38°C;

$t$  = actual temperature of the substance during filling (in °C) (when this temperature is unknown, let  $t = 15^\circ\text{C}$ );

The value of  $F$  given above for insulated shells may be taken provided that the insulation is in conformance with 6.7.2.12.2.4;



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- A** = total external surface area of shell in square metres;  
**Z** = the gas compressibility factor in the accumulating condition (when this factor is unknown, let *Z* equal 1.0);  
**T** = absolute temperature in kelvin (°C + 273) above the pressure relief devices in the accumulating condition;  
**L** = the latent heat of vaporization of the liquid, in kJ/kg, in the accumulating condition;  
**M** = molecular mass of the discharged gas;  
**C** = a constant which is derived from one of the following formulae as a function of the ratio *k* of specific heats:

$$k = \frac{C_p}{C_v}$$

where:

$C_p$  = specific heat at constant pressure; and

$C_v$  = specific heat at constant volume.

When  $k > 1$ :

$$C = \sqrt{k \left( \frac{2}{k+1} \right)^{\frac{k+1}{k-1}}}$$

When  $k = 1$  or  $k$  is unknown:

$$C = \frac{1}{\sqrt{e}} = 0.607$$

where *e* is the mathematical constant 2.7183.

*C* may also be taken from the following table:

| <i>k</i> | <i>C</i> | <i>k</i> | <i>C</i> | <i>k</i> | <i>C</i> |
|----------|----------|----------|----------|----------|----------|
| 1.00     | 0.607    | 1.26     | 0.660    | 1.52     | 0.704    |
| 1.02     | 0.611    | 1.28     | 0.664    | 1.54     | 0.707    |
| 1.04     | 0.615    | 1.30     | 0.667    | 1.56     | 0.71     |
| 1.06     | 0.620    | 1.32     | 0.671    | 1.58     | 0.713    |
| 1.08     | 0.624    | 1.34     | 0.674    | 1.60     | 0.716    |
| 1.10     | 0.628    | 1.36     | 0.678    | 1.62     | 0.719    |
| 1.12     | 0.633    | 1.38     | 0.681    | 1.64     | 0.722    |
| 1.14     | 0.637    | 1.40     | 0.685    | 1.66     | 0.725    |
| 1.16     | 0.641    | 1.42     | 0.688    | 1.68     | 0.728    |
| 1.18     | 0.645    | 1.44     | 0.691    | 1.70     | 0.731    |
| 1.20     | 0.649    | 1.46     | 0.695    | 2.0      | 0.77     |
| 1.22     | 0.652    | 1.48     | 0.698    | 2.2      | 0.793    |
| 1.24     | 0.656    | 1.50     | 0.701    |          |          |

**6.7.2.12.2.2** As an alternative to the formula above, shells designed for the transport of liquids may have their relief devices sized in accordance with the table in 6.7.2.12.2.3. This table assumes an insulation value of  $F = 1$  and shall be adjusted accordingly when the shell is insulated. Other values used in determining this table are:

$$M = 86.7; \quad T = 394 \text{ K}; \quad L = 334.94 \text{ kJ/kg}; \quad C = 0.607; \quad Z = 1$$

**6.7.2.12.2.3** Minimum required rate of discharge, *Q*, in cubic metres of air per second at 1 bar and 0°C (273 K):

| <b>A</b><br>Exposed area<br>(square metres) | <b>Q</b><br>(cubic metres of<br>air per second) | <b>A</b><br>Exposed area<br>(square metres) | <b>Q</b><br>(cubic metres of<br>air per second) |
|---------------------------------------------|-------------------------------------------------|---------------------------------------------|-------------------------------------------------|
| 2                                           | 0.230                                           | 14                                          | 1.132                                           |
| 3                                           | 0.320                                           | 16                                          | 1.263                                           |
| 4                                           | 0.405                                           | 18                                          | 1.391                                           |
| 5                                           | 0.487                                           | 20                                          | 1.517                                           |
| 6                                           | 0.565                                           | 22.5                                        | 1.670                                           |
| 7                                           | 0.641                                           | 25                                          | 1.821                                           |
| 8                                           | 0.715                                           | 27.5                                        | 1.969                                           |
| 9                                           | 0.788                                           | 30                                          | 2.115                                           |
| 10                                          | 0.859                                           | 32.5                                        | 2.258                                           |
| 12                                          | 0.998                                           | 35                                          | 2.400                                           |

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| A<br>Exposed area<br>(square metres) | Q<br>(cubic metres of<br>air per second) | A<br>Exposed area<br>(square metres) | Q<br>(cubic metres of<br>air per second) |
|--------------------------------------|------------------------------------------|--------------------------------------|------------------------------------------|
| 37.5                                 | 2.539                                    | 62.5                                 | 3.860                                    |
| 40                                   | 2.677                                    | 65                                   | 3.987                                    |
| 42.5                                 | 2.814                                    | 67.5                                 | 4.112                                    |
| 45                                   | 2.949                                    | 70                                   | 4.236                                    |
| 47.5                                 | 3.082                                    | 75                                   | 4.483                                    |
| 50                                   | 3.215                                    | 80                                   | 4.726                                    |
| 52.5                                 | 3.346                                    | 85                                   | 4.967                                    |
| 55                                   | 3.476                                    | 90                                   | 5.206                                    |
| 57.5                                 | 3.605                                    | 95                                   | 5.442                                    |
| 60                                   | 3.733                                    | 100                                  | 5.676                                    |

6.7.2.12.2.4 Insulation systems, used for the purpose of reducing venting capacity, shall be approved by the competent authority or its authorized body. In all cases, insulation systems approved for this purpose shall:

- (a) remain effective at all temperatures up to 649°C; and
- (b) be jacketed with a material having a melting point of 700°C or greater.

#### 6.7.2.13 Marking of pressure relief devices

6.7.2.13.1 Every pressure relief device shall be clearly and permanently marked with the following:

- .1 the pressure (in bar or kPa) or temperature (in °C) at which it is set to discharge;
- .2 the allowable tolerance at the discharge pressure, for spring-loaded devices;
- .3 the reference temperature corresponding to the rated pressure, for frangible discs;
- .4 the allowable temperature tolerance, for fusible elements;
- .5 the rated flow capacity of the spring-loaded pressure relief devices, frangible discs or fusible elements in standard cubic metres of air per second (m<sup>3</sup>/s); and
- .6 the cross sectional flow areas of the spring-loaded pressure relief devices, frangible discs and fusible elements in mm<sup>2</sup>.

When practicable, the following information shall also be shown:

- .7 the manufacturer's name and relevant catalogue number.

6.7.2.13.2 The rated flow capacity marked on the spring-loaded pressure relief devices shall be determined according to ISO 4126-1:2004 and ISO 4126-7:2004.

#### 6.7.2.14 Connections to pressure relief devices

6.7.2.14.1 Connections to pressure relief devices shall be of sufficient size to enable the required discharge to pass unrestricted to the safety device. No stop-valve shall be installed between the shell and the pressure relief devices except where duplicate devices are provided for maintenance or other reasons and the stop-valves serving the devices actually in use are locked open or the stop-valves are interlocked so that at least one of the duplicate devices is always in use. There shall be no obstruction in an opening leading to a vent or pressure relief device which might restrict or cut off the flow from the shell to that device. Vents or pipes from the pressure relief device outlets, when used, shall deliver the relieved vapour or liquid to the atmosphere in conditions of minimum back-pressure on the relieving devices.

#### 6.7.2.15 Siting of pressure relief devices

6.7.2.15.1 Each pressure relief device inlet shall be situated on top of the shell in a position as near the longitudinal and transverse centre of the shell as reasonably practicable. All pressure relief device inlets shall, under maximum filling conditions, be situated in the vapour space of the shell and the devices shall be so arranged as to ensure the escaping vapour is discharged unrestrictedly. For flammable substances, the escaping vapour shall be directed away from the shell in such a manner that it cannot impinge upon the shell. Protective devices which deflect the flow of vapour are permissible provided the required relief-device capacity is not reduced.

6.7.2.15.2 Arrangements shall be made to prevent access to the pressure relief devices by unauthorized persons and to protect the devices from damage caused by the portable tank overturning.

**Part 6 – Construction and testing of packagings, IBCs, etc.****6.7.2.16 Gauging devices**

- 6.7.2.16.1 Glass level-gauges and gauges made of other fragile material, which are in direct communication with the contents of the tank, shall not be used.

**6.7.2.17 Portable tank supports, frameworks, lifting and tie-down attachments**

- 6.7.2.17.1 Portable tanks shall be designed and constructed with a support structure to provide a secure base during transport. The forces specified in 6.7.2.2.12 and the safety factor specified in 6.7.2.2.13 shall be considered in this aspect of the design. Skids, frameworks, cradles or other similar structures are acceptable.

- 6.7.2.17.2 The combined stresses caused by portable tank mountings (such as cradles, framework, etc.) and portable tank lifting and tie-down attachments shall not cause excessive stress in any portion of the shell. Permanent lifting and tie-down attachments shall be fitted to all portable tanks. Preferably they shall be fitted to the portable tank supports but may be secured to reinforcing plates located on the shell at the points of support.

- 6.7.2.17.3 In the design of supports and frameworks, the effects of environmental corrosion shall be taken into account.

- 6.7.2.17.4 Forklift pockets shall be capable of being closed off. The means of closing forklift pockets shall be a permanent part of the framework or permanently attached to the framework. Single-compartment portable tanks with a length less than 3.65 m need not have closed-off forklift pockets provided that:

- .1 the shell, including all the fittings, is well protected from being hit by the forklift blades; and
- .2 the distance between the centres of the forklift pockets is at least half of the maximum length of the portable tank.

- 6.7.2.17.5 When portable tanks are not protected during transport, according to 4.2.1.2, the shells and service equipment shall be protected against damage to the shell and service equipment resulting from lateral or longitudinal impact or overturning. External fittings shall be protected so as to preclude the release of the shell contents upon impact or overturning of the portable tank on its fittings. Examples of protection include:

- .1 protection against lateral impact, which may consist of longitudinal bars protecting the shell on both sides at the level of the median line;
- .2 protection of the portable tank against overturning, which may consist of reinforcement rings or bars fixed across the frame;
- .3 protection against rear impact, which may consist of a bumper or frame;
- .4 protection of the shell against damage from impact or overturning by use of an ISO frame in accordance with ISO 1496-3:1995.

**6.7.2.18 Design approval**

- 6.7.2.18.1 The competent authority or its authorized body shall issue a design approval certificate for any new design of a portable tank. This certificate shall attest that a portable tank has been surveyed by that authority, is suitable for its intended purpose and meets the provisions of this chapter and, where appropriate, the provisions for substances provided in chapter 4.2 and in the Dangerous Goods List in chapter 3.2. When a series of portable tanks are manufactured without change in the design, the certificate shall be valid for the entire series. The certificate shall refer to the prototype test report, the substances or group of substances allowed to be transported, the materials of construction of the shell and lining (when applicable) and an approval number. The approval number shall consist of the distinguishing sign or mark of the State in whose territory the approval was granted, indicated by the distinguishing sign used on vehicles in international road traffic\* and a registration number. Any alternative arrangements according to 6.7.1.2 shall be indicated on the certificate. A design approval may serve for the approval of smaller portable tanks made of materials of the same kind and thickness, by the same fabrication techniques and with identical supports, equivalent closures and other appurtenances.

- 6.7.2.18.2 The prototype test report for the design approval shall include at least the following:

- .1 the results of the applicable framework test specified in ISO 1496-3:1995;
- .2 the results of the initial inspection and test in 6.7.2.19.3; and
- .3 the results of the impact test in 6.7.2.19.1, when applicable.

**6.7.2.19 Inspection and testing**

- 6.7.2.19.1 Portable tanks meeting the definition of *container* in the *International Convention for Safe Containers, 1972* (CSC Convention), as amended, shall not be used unless they are successfully qualified by subjecting a representative prototype of each design to the Dynamic, Longitudinal Impact Test prescribed in the *Manual*

\* Distinguishing sign of the State of registration used on motor vehicles and trailers in international road traffic, e.g. in accordance with the Geneva Convention on Road Traffic of 1949 or the Vienna Convention on Road Traffic of 1968.

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of *Tests and Criteria*, part IV, section 41. This provision only applies to portable tanks which are constructed according to a design approval certificate which has been issued on or after 1 January 2008.

- 6.7.2.19.2** The shell and items of equipment of each portable tank shall be inspected and tested before being put into service for the first time (initial inspection and test) and thereafter at not more than five-year intervals (5-year periodic inspection and test) with an intermediate periodic inspection and test (2.5-year periodic inspection and test) midway between the 5-year periodic inspections and tests. The 2.5-year periodic inspection and test may be performed within 3 months of the specified date. An exceptional inspection and test shall be performed regardless of the date of the last periodic inspection and test when necessary according to 6.7.2.19.7.
- 6.7.2.19.3** The initial inspection and test of a portable tank shall include a check of the design characteristics, an internal and external examination of the portable tank and its fittings with due regard to the substances to be transported, and a pressure test. Before the portable tank is placed into service, a leakproofness test and a test of the satisfactory operation of all service equipment shall also be performed. When the shell and its fittings have been pressure-tested separately, they shall be subjected together after assembly to a leakproofness test.
- 6.7.2.19.4** The 5-year periodic inspection and test shall include an internal and external examination and, as a general rule, a hydraulic pressure test. For tanks only used for the transport of solid substances other than toxic or corrosive substances, which do not liquefy during transport, the hydraulic pressure test may be replaced by a suitable pressure test at 1.5 times MAWP, subject to competent authority approval. Sheathing, thermal insulation and the like shall be removed only to the extent required for reliable appraisal of the condition of the portable tank. When the shell and equipment have been pressure-tested separately, they shall be subjected together after assembly to a leakproofness test.
- 6.7.2.19.4.1** The heating system shall be subject to inspection and tests including pressure tests on heating coils or ducts during the 5-year periodic inspection.
- 6.7.2.19.5** The intermediate 2.5-year periodic inspection and test shall at least include an internal and external examination of the portable tank and its fittings with due regard to the substances intended to be transported, a leakproofness test and a test of the satisfactory operation of all service equipment. Sheathing, thermal insulation and the like shall be removed only to the extent required for reliable appraisal of the condition of the portable tank. For portable tanks dedicated to the transport of a single substance, the 2.5-year internal examination may be waived or substituted by other test methods or inspection procedures specified by the competent authority or its authorized body.
- 6.7.2.19.6** A portable tank may not be filled and offered for transport after the date of expiry of the last 5-year or 2.5-year periodic inspection and test as required by 6.7.2.19.2. However, a portable tank filled prior to the date of expiry of the last periodic inspection and test may be transported for a period not to exceed three months beyond the date of expiry of the last periodic test or inspection. In addition, a portable tank may be transported after the date of expiry of the last periodic test and inspection:
- .1 after emptying but before cleaning, for purposes of performing the next required test or inspection prior to refilling; and
  - .2 unless otherwise approved by the competent authority, for a period not to exceed six months beyond the date of expiry of the last periodic test or inspection, in order to allow the return of dangerous goods for proper disposal or recycling. Reference to this exemption shall be mentioned in the transport document.
- Except as provided for in this paragraph, portable tanks which have missed the timeframe for their scheduled 5-year or 2.5-year periodic inspection and test may only be filled and offered for transport if a new 5-year periodic inspection and test is performed according to 6.7.2.19.4.
- 6.7.2.19.7** The exceptional inspection and test is necessary when the portable tank shows evidence of damaged or corroded areas, or leakage, or other conditions that indicate a deficiency that could affect the integrity of the portable tank. The extent of the exceptional inspection and test shall depend on the amount of damage or deterioration of the portable tank. It shall include at least the 2.5-year periodic inspection and test according to 6.7.2.19.5.
- 6.7.2.19.8** The internal and external examinations shall ensure that:
- .1 the shell is inspected for pitting, corrosion, or abrasions, dents, distortions, defects in welds or any other conditions, including leakage, that might render the portable tank unsafe for transport. The wall thickness shall be verified by appropriate measurement if this inspection indicates a reduction of wall thickness;
  - .2 the piping, valves, heating/cooling system, and gaskets are inspected for corroded areas, defects, or any other conditions, including leakage, that might render the portable tank unsafe for filling, discharge or transport;
  - .3 devices for tightening manhole covers are operative and there is no leakage at manhole covers or gaskets;
  - .4 missing or loose bolts or nuts on any flanged connection or blank flange are replaced or tightened;

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- .5 all emergency devices and valves are free from corrosion, distortion and any damage or defect that could prevent their normal operation. Remote closure devices and self-closing stop-valves shall be operated to demonstrate proper operation;
- .6 linings, if any, are inspected in accordance with criteria outlined by the lining manufacturer;
- .7 required marks on the portable tank are legible and in accordance with the applicable provisions; and
- .8 the framework, supports and arrangements for lifting the portable tank are in a satisfactory condition.

**6.7.2.19.9** The inspections and tests in 6.7.2.19.1, 6.7.2.19.3, 6.7.2.19.4, 6.7.2.19.5 and 6.7.2.19.7 shall be performed or witnessed by an expert approved by the competent authority or its authorized body. When the pressure test is a part of the inspection and test, the test pressure shall be the one indicated on the data plate of the portable tank. While under pressure, the portable tank shall be inspected for any leaks in the shell, piping or equipment.

**6.7.2.19.10** In all cases when cutting, burning or welding operations on the shell have been effected, that work shall be to the approval of the competent authority or its authorized body, taking into account the pressure-vessel code used for the construction of the shell. A pressure test to the original test pressure shall be performed after the work is completed.

**6.7.2.19.11** When evidence of any unsafe condition is discovered, the portable tank shall not be returned to service until it has been corrected and the test is repeated and passed.

**6.7.2.20 Marking**

**6.7.2.20.1** Every portable tank shall be fitted with a corrosion-resistant metal plate permanently attached to the portable tank in a conspicuous place readily accessible for inspection. When for reasons of portable tank arrangements the plate cannot be permanently attached to the shell, the shell shall be marked with at least the information required by the pressure-vessel code. As a minimum, at least the following information shall be marked on the plate by stamping or by any other similar method:

- (a) Owner information
  - (i) Owner's registration number;
- (b) Manufacturing information
  - (i) Country of manufacture;
  - (ii) Year of manufacture;
  - (iii) Manufacturer's name or mark;
  - (iv) Manufacturer's serial number;
- (c) Approval information
  - (i) The United Nations packaging symbol:



This symbol shall not be used for any purpose other than certifying that a packaging, a flexible bulk container, a portable tank or a MEGC complies with the relevant requirements in chapters 6.1, 6.2, 6.3, 6.5, 6.6, 6.7 or 6.9;

- (ii) Approval country;
- (iii) Authorized body for the design approval;
- (iv) Design approval number;
- (v) Letters "AA", if the design was approved under alternative arrangements (see 6.7.1.2);
- (vi) Pressure-vessel code to which the shell is designed;
- (d) Pressures
  - (i) MAWP (in bar gauge or kPa gauge);\*
  - (ii) Test pressure (in bar gauge or kPa gauge);\*
  - (iii) Initial pressure test date (month and year);
  - (iv) Identification mark of the initial pressure test witness;
  - (v) External design pressure<sup>†</sup> (in bar gauge or kPa gauge);\*
  - (vi) MAWP for heating/cooling system (in bar gauge or kPa gauge)\* (when applicable);
- (e) Temperatures
  - (i) Design temperature range (in °C);\*

\* The unit used shall be indicated.

<sup>†</sup> See 6.7.2.2.10.



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- (f) Materials
  - (i) Shell material(s) and material standard reference(s);
  - (ii) Equivalent thickness in reference steel (in mm);\* and
  - (iii) Lining material (when applicable);
- (g) Capacity
  - (i) Tank water capacity at 20°C (in litres);\*  
This indication is to be followed by the symbol “S” when the shell is divided by surge plates into sections of not more than 7,500 L capacity;
  - (ii) Water capacity of each compartment at 20°C (in litres)\* (when applicable, for multi-compartment tanks).  
This indication is to be followed by the symbol “S” when the compartment is divided by surge plates into sections of not more than 7,500 L capacity;
- (h) Periodic inspections and tests
  - (i) Type of the most recent periodic test (2.5-year, 5-year or exceptional);
  - (ii) Date of the most recent periodic test (month and year);
  - (iii) Test pressure (in bar gauge or kPa gauge)\* of the most recent periodic test (if applicable);
  - (iv) Identification mark of the authorized body who performed or witnessed the most recent test.

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\* The unit used shall be indicated.

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Figure 6.7.2.20.1 – Example of a plate for marking

|                                                                                              |                                     |                |                      |
|----------------------------------------------------------------------------------------------|-------------------------------------|----------------|----------------------|
| Owner's registration number                                                                  |                                     |                |                      |
| <b>MANUFACTURING INFORMATION</b>                                                             |                                     |                |                      |
| Country of manufacture                                                                       |                                     |                |                      |
| Year of manufacture                                                                          |                                     |                |                      |
| Manufacturer                                                                                 |                                     |                |                      |
| Manufacturer's serial number                                                                 |                                     |                |                      |
| <b>APPROVAL INFORMATION</b>                                                                  |                                     |                |                      |
|             | Approval country                    |                |                      |
|                                                                                              | Authorized body for design approval |                |                      |
|                                                                                              | Design approval number              |                | "AA" (if applicable) |
| Shell design code (pressure-vessel code)                                                     |                                     |                |                      |
| <b>PRESSURES</b>                                                                             |                                     |                |                      |
| MAWP                                                                                         |                                     | bar or kPa     |                      |
| Test pressure                                                                                |                                     | bar or kPa     |                      |
| Initial pressure test date:                                                                  | (mm/yyyy)                           | Witness stamp: |                      |
| External design pressure                                                                     |                                     | bar or kPa     |                      |
| MAWP for heating/cooling system (when applicable)                                            |                                     | bar or kPa     |                      |
| <b>TEMPERATURES</b>                                                                          |                                     |                |                      |
| Design temperature range                                                                     |                                     | °C to °C       |                      |
| <b>MATERIALS</b>                                                                             |                                     |                |                      |
| Shell material(s) and material standard reference(s)                                         |                                     |                |                      |
| Equivalent thickness in reference steel                                                      |                                     | mm             |                      |
| Lining material (when applicable)                                                            |                                     |                |                      |
| <b>CAPACITY</b>                                                                              |                                     |                |                      |
| Tank water capacity at 20°C                                                                  |                                     | litres         | "S" (if applicable)  |
| Water capacity of compartment ____ at 20°C<br>(when applicable, for multi-compartment tanks) |                                     | litres         | "S" (if applicable)  |

| PERIODIC INSPECTIONS/TESTS |           |                                              |            |           |           |                                              |            |
|----------------------------|-----------|----------------------------------------------|------------|-----------|-----------|----------------------------------------------|------------|
| Test type                  | Test date | Witness stamp and test pressure <sup>a</sup> |            | Test type | Test date | Witness stamp and test pressure <sup>a</sup> |            |
|                            | (mm/yyyy) |                                              | bar or kPa |           | (mm/yyyy) |                                              | bar or kPa |
|                            |           |                                              |            |           |           |                                              |            |
|                            |           |                                              |            |           |           |                                              |            |
|                            |           |                                              |            |           |           |                                              |            |
|                            |           |                                              |            |           |           |                                              |            |

<sup>a</sup> Test pressure if applicable.

## Chapter 6.7 – Design, construction, inspection and testing of portable tanks &amp; MEGCs

- 6.7.2.20.2 The following information shall be durably marked either on the portable tank itself or on a metal plate firmly secured to the portable tank:

Name of the operator  
Maximum permissible gross mass (MPGM) . . . . . kg  
Unladen (tare) mass . . . . . kg  
Portable tank instruction in accordance with 4.2.5.2.6.

- 6.7.2.20.3 If a portable tank is designed and approved for handling in open seas, the words “OFFSHORE PORTABLE TANK” shall be marked on the identification plate.

### 6.7.3 Provisions for the design, construction, inspection and testing of portable tanks intended for the transport of non-refrigerated liquefied gases of class 2

**Note:** These requirements also apply to portable tanks intended for the transport of chemicals under pressure (UN Nos. 3500, 3501, 3502, 3503, 3504 and 3505).

#### 6.7.3.1 Definitions

For the purposes of this section:

*Design pressure* means the pressure to be used in calculations required by a recognized pressure-vessel code. The design pressure shall be not less than the highest of the following pressures:

- .1 the maximum effective gauge pressure allowed in the shell during filling or discharge; or
- .2 the sum of:
  - .1 the maximum effective gauge pressure to which the shell is designed, as defined in .2 of the MAWP definition (see below); and
  - .2 a head pressure determined on the basis of the static forces specified in 6.7.3.2.9, but not less than 0.35 bar;

*Design reference temperature* means the temperature at which the vapour pressure of the contents is determined for the purpose of calculating the MAWP. The design reference temperature shall be less than the critical temperature of the non-refrigerated liquefied gas or liquefied gas propellants of chemicals under pressure intended to be transported to ensure that the gas at all times is liquefied. This value for each portable tank type is as follows:

- .1 shell with a diameter of 1.5 m or less: 65°C;
- .2 shell with a diameter of more than 1.5 m:
  - .1 without insulation or sunshield: 60°C;
  - .2 with sunshield (see 6.7.3.2.12): 55°C; and
  - .3 with insulation (see 6.7.3.2.12): 50°C;

*Design temperature range* for the shell shall be –40°C to 50°C for non-refrigerated liquefied gases transported under ambient conditions. More severe design temperatures shall be considered for portable tanks subjected to severe climatic conditions;

*Filling density* means the average mass of non-refrigerated liquefied gas per litre of shell capacity (kg/L). The filling density is given in portable tank instruction T50 in 4.2.5.2.6;

*Leakproofness test* means a test using gas subjecting the shell and its service equipment to an effective internal pressure of not less than 25% of the MAWP;

*Maximum allowable working pressure* (MAWP) means a pressure that shall be not less than the highest of the following pressures measured at the top of the shell while in operating position, but in no case less than 7 bar:

- .1 the maximum effective gauge pressure allowed in the shell during filling or discharge; or
- .2 the maximum effective gauge pressure to which the shell is designed, which shall be:
  - .1 for a non-refrigerated liquefied gas listed in the portable tank instruction T50 in 4.2.5.2.6, the MAWP (in bar) given in portable tank instruction T50 for that gas;
  - .2 for other non-refrigerated liquefied gases, not less than the sum of:
    - the absolute vapour pressure (in bar) of the non-refrigerated liquefied gas at the design reference temperature minus 1 bar; and
    - the partial pressure (in bar) of air or other gases in the ullage space, being determined by the design reference temperature and the liquid phase expansion due to an increase of the mean bulk temperature of  $t_r - t_i$  ( $t_i$  = filling temperature, usually 15°C;  $t_r$  = 50°C, maximum mean bulk temperature);

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- .3 for chemicals under pressure, the MAWP (in bar) given in T50 portable tank instruction for the liquefied gas portion of the propellants listed in T50 in 4.2.5.2.6.

*Maximum permissible gross mass (MPGM)* means the sum of the tare mass of the portable tank and the heaviest load authorized for transport;

*Mild steel* means a steel with a guaranteed minimum tensile strength of 360 N/mm<sup>2</sup> to 440 N/mm<sup>2</sup> and a guaranteed minimum elongation at fracture conforming to 6.7.3.3.3.3;

*Portable tank* means a multimodal tank having a capacity of more than 450 L used for the transport of non-refrigerated liquefied gases of class 2. The portable tank includes a shell fitted with service equipment and structural equipment necessary for the transport of gases. The portable tank shall be capable of being filled and discharged without the removal of its structural equipment. It shall possess stabilizing members external to the shell, and shall be capable of being lifted when full. It shall be designed primarily to be loaded onto a transport vehicle or ship and shall be equipped with skids, mountings or accessories to facilitate mechanical handling. Road tank-vehicles, rail tank-wagons, non-metallic tanks, intermediate bulk containers (IBCs), gas cylinders and large receptacles are not considered to fall within the definition for portable tanks;

*Reference steel* means a steel with a tensile strength of 370 N/mm<sup>2</sup> and an elongation at fracture of 27%;

*Service equipment* means measuring instruments and filling, discharge, venting, safety and insulating devices;

*Shell* means the part of the portable tank which retains the non-refrigerated liquefied gas intended for transport (tank proper), including openings and their closures, but does not include service equipment or external structural equipment;

*Structural equipment* means reinforcing, fastening, protective and stabilizing members external to the shell;

*Test pressure* means the maximum gauge pressure at the top of the shell during the pressure test.

### 6.7.3.2 General design and construction provisions

- 6.7.3.2.1 Shells shall be designed and constructed in accordance with the provisions of a pressure-vessel code recognized by the competent authority. Shells shall be made of steel suitable for forming. The materials shall, in principle, conform to national or international material standards. For welded shells, only a material whose weldability has been fully demonstrated shall be used. Welds shall be skilfully made and afford complete safety. When the manufacturing process or the materials make it necessary, the shells shall be suitably heat-treated to guarantee adequate toughness in the weld and in the heat-affected zones. In choosing the material, the design temperature range shall be taken into account with respect to risk of brittle fracture, to stress corrosion cracking and to resistance to impact. When fine-grain steel is used, the guaranteed value of the yield strength shall be not more than 460 N/mm<sup>2</sup> and the guaranteed value of the upper limit of the tensile strength shall be not more than 725 N/mm<sup>2</sup>, according to the material specification. Portable tank materials shall be suitable for the external environment in which they may be transported.
- 6.7.3.2.2 Portable tank shells, fittings and pipework shall be constructed of materials which are:
- .1 substantially immune to attack by the non-refrigerated liquefied gas(es) intended to be transported; or
  - .2 properly passivated or neutralized by chemical reaction.
- 6.7.3.2.3 Gaskets shall be made of materials compatible with the non-refrigerated liquefied gas(es) intended to be transported.
- 6.7.3.2.4 Contact between dissimilar metals which could result in damage by galvanic action shall be avoided.
- 6.7.3.2.5 The materials of the portable tank, including any devices, gaskets, and accessories, shall not adversely affect the non-refrigerated liquefied gas(es) intended for transport in the portable tank.
- 6.7.3.2.6 Portable tanks shall be designed and constructed with supports to provide a secure base during transport and with suitable lifting and tie-down attachments.
- 6.7.3.2.7 Portable tanks shall be designed to withstand, without loss of contents, at least the internal pressure due to the contents and the static, dynamic and thermal loads during normal conditions of handling and transport. The design shall demonstrate that the effects of fatigue, caused by repeated application of these loads through the expected life of the portable tank, have been taken into account.
- 6.7.3.2.7.1 For portable tanks that are intended for use as offshore tank-containers, the dynamic stresses imposed by handling in open seas shall be taken into account.
- 6.7.3.2.8 Shells shall be designed to withstand an external pressure of at least 0.4 bar gauge above the internal pressure without permanent deformation. When the shell is to be subjected to a significant vacuum before filling or during discharge, it shall be designed to withstand an external pressure of at least 0.9 bar gauge above the internal pressure and shall be proven at that pressure.

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- 6.7.3.2.9 Portable tanks and their fastenings shall, under the maximum permissible load, be capable of absorbing the following separately applied static forces:
- .1 in the direction of travel: twice the MPGM multiplied by the acceleration due to gravity (g);\*
  - .2 horizontally at right angles to the direction of travel: the MPGM (when the direction of travel is not clearly determined, the forces shall be equal to twice the MPGM) multiplied by the acceleration due to gravity (g);\*
  - .3 vertically upwards: the MPGM multiplied by the acceleration due to gravity (g);\* and
  - .4 vertically downwards: twice the MPGM (total loading including the effect of gravity) multiplied by the acceleration due to gravity (g).\*
- 6.7.3.2.10 Under each of the forces in 6.7.3.2.9, the safety factor to be observed shall be as follows:
- .1 for steels having a clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed yield strength; or
  - .2 for steels with no clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed 0.2% proof strength and, for austenitic steels, the 1% proof strength.
- 6.7.3.2.11 The values of yield strength or proof strength shall be the values according to national or international material standards. When austenitic steels are used, the specified minimum values of yield strength and proof strength according to the material standards may be increased by up to 15% when these greater values are attested in the material inspection certificate. When no material standard exists for the steel in question, the value of yield strength or proof strength used shall be approved by the competent authority.
- 6.7.3.2.12 When the shells intended for the transport of non-refrigerated liquefied gases are equipped with thermal insulation, the thermal insulation system shall satisfy the following provisions:
- .1 It shall consist of a shield covering not less than the upper third but not more than the upper half of the surface of the shell and separated from the shell by an air space about 40 mm across; or
  - .2 It shall consist of a complete cladding of adequate thickness of insulating materials, protected so as to prevent the ingress of moisture and damage under normal conditions of transport and so as to provide a thermal conductance of not more than 0.67 W/m·K;
  - .3 When the protective covering is so closed as to be gas-tight, a device shall be provided to prevent any dangerous pressure from developing in the insulating layer in the event of inadequate gas-tightness of the shell or of its items of equipment;
  - .4 The thermal insulation shall not inhibit access to the fittings and discharge devices.
- 6.7.3.2.13 Portable tanks intended for the transport of flammable non-refrigerated liquefied gases shall be capable of being electrically earthed.
- 6.7.3.3 Design criteria**
- 6.7.3.3.1 Shells shall be of a circular cross-section.
- 6.7.3.3.2 Shells shall be designed and constructed to withstand a test pressure not less than 1.3 times the design pressure. The shell design shall take into account the minimum MAWP values provided in portable tank instruction T50 in 4.2.5.2.6 for each non-refrigerated liquefied gas intended for transport. Attention is drawn to the minimum shell thickness provisions for these shells specified in 6.7.3.4.
- 6.7.3.3.3 For steels exhibiting a clearly defined yield point or characterized by a guaranteed proof strength (0.2% proof strength, generally, or 1% proof strength for austenitic steels), the primary membrane stress  $\sigma$  (sigma) in the shell shall not exceed  $0.75R_e$  or  $0.50R_m$ , whichever is lower, at the test pressure, where:
- $$R_e = \text{yield strength in N/mm}^2, \text{ or } 0.2\% \text{ proof strength or, for austenitic steels, } 1\% \text{ proof strength.}$$
- $$R_m = \text{minimum tensile strength in N/mm}^2.$$
- 6.7.3.3.3.1 The values of  $R_e$  and  $R_m$  to be used shall be the specified minimum values according to national or international material standards. When austenitic steels are used, these specified minimum values for  $R_e$  and  $R_m$  according to the material standards may be increased by up to 15% when these greater values are attested in the material inspection certificate. When no material standard exists for the steel in question, the values of  $R_e$  and  $R_m$  used shall be approved by the competent authority or its authorized body.
- 6.7.3.3.3.2 Steels which have a  $R_e/R_m$  ratio of more than 0.85 are not allowed for the construction of welded shells. The values of  $R_e$  and  $R_m$  to be used in determining this ratio shall be the values specified in the material inspection certificate.
- 6.7.3.3.3.3 Steels used in the construction of shells shall have an elongation at fracture, in %, of not less than  $10,000/R_m$  with an absolute minimum of 16% for fine-grain steels and 20% for other steels.
- 6.7.3.3.3.4 For the purpose of determining actual values for materials, it shall be noted that for sheet metal, the axis of the tensile test specimen shall be at right angles (transversely) to the direction of rolling. The permanent

\* For calculation purposes,  $g = 9.81 \text{ m/s}^2$ .



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elongation at fracture shall be measured on test specimens of rectangular cross-section in accordance with ISO 6892:1998 using a 50 mm gauge length.

**6.7.3.4 Minimum shell thickness**

**6.7.3.4.1** The minimum shell thickness shall be the greater thickness based on:

- .1 the minimum thickness determined in accordance with the provisions in 6.7.3.4; and
- .2 the minimum thickness determined in accordance with the recognized pressure-vessel code, including the provisions in 6.7.3.3.

In addition, any relevant portable tank special provision indicated in column 14 of the Dangerous Goods List and described in 4.2.5.3 shall be taken into account.

**6.7.3.4.2** The cylindrical portions, ends (heads) and manhole covers of shells of not more than 1.80 m in diameter shall be not less than 5 mm thick in the reference steel or of equivalent thickness in the steel to be used. Shells of more than 1.80 m in diameter shall be not less than 6 mm thick in the reference steel or of equivalent thickness in the steel to be used.

**6.7.3.4.3** The cylindrical portions, ends (heads) and manhole covers of all shells shall be not less than 4 mm thick regardless of the material of construction.

**6.7.3.4.4** The equivalent thickness of a steel other than the thickness prescribed for the reference steel in 6.7.3.4.2 shall be determined using the following formula:

$$e_1 = \frac{21.4 \times e_0}{\sqrt[3]{R_{m1} \times A_1}}$$

where:

- $e_1$  = required equivalent thickness (in mm) of the steel to be used;
- $e_0$  = minimum thickness (in mm) of the reference steel specified in 6.7.3.4.2;
- $R_{m1}$  = guaranteed minimum tensile strength (in N/mm<sup>2</sup>) of the steel to be used (see 6.7.3.3.3);
- $A_1$  = guaranteed minimum elongation at fracture (in %) of the steel to be used according to national or international standards.

**6.7.3.4.5** In no case shall the wall thickness be less than that prescribed in 6.7.3.4.1 to 6.7.3.4.3. All parts of the shell shall have a minimum thickness as determined by 6.7.3.4.1 to 6.7.3.4.3. This thickness shall be exclusive of any corrosion allowance.

**6.7.3.4.6** When mild steel is used (see 6.7.3.1), calculation using the equation in 6.7.3.4.4 is not required.

**6.7.3.4.7** There shall be no sudden change of plate thickness at the attachment of the ends (heads) to the cylindrical portion of the shell.

**6.7.3.5 Service equipment**

**6.7.3.5.1** Service equipment shall be so arranged as to be protected against the risk of being wrenched off or damaged during handling and transport. When the connection between the frame and the shell allows relative movement between the sub-assemblies, the equipment shall be so fastened as to permit such movement without risk of damage to working parts. The external discharge fittings (pipe sockets, shut-off devices), the internal stop-valve and its seating shall be protected against the danger of being wrenched off by external forces (for example, by using shear sections). The filling and discharge devices (including flanges or threaded plugs) and any protective caps shall be capable of being secured against unintended opening.

**6.7.3.5.1.1** For offshore tank-containers, where positioning of service equipment and the design and strength of protection for such equipment is concerned, the increased danger of impact damage when handling such tanks in open seas shall be taken into account.

**6.7.3.5.2** All openings with a diameter of more than 1.5 mm in shells of portable tanks, except openings for pressure relief devices, inspection openings and closed bleed holes, shall be fitted with at least three mutually independent shut-off devices in series, the first being an internal stop-valve, excess flow valve or equivalent device, the second being an external stop-valve and the third being a blank flange or equivalent device.

**6.7.3.5.2.1** When a portable tank is fitted with an excess flow valve, the excess flow valve shall be so fitted that its seating is inside the shell or inside a welded flange or, when fitted externally, its mountings shall be designed so that, in the event of impact, its effectiveness shall be maintained. The excess flow valves shall be selected and fitted so as to close automatically when the rated flow specified by the manufacturer is reached. Connections and accessories leading to or from such a valve shall have a capacity for a flow more than the rated flow of the excess flow valve.

**6.7.3.5.3** For filling and discharge openings, the first shut-off device shall be an internal stop-valve and the second shall be a stop-valve placed in an accessible position on each discharge and filling pipe.

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- 6.7.3.5.4 For filling and discharge bottom openings of portable tanks intended for the transport of flammable and/or toxic non-refrigerated liquefied gases or chemicals under pressure, the internal stop-valve shall be a quick-closing safety device which closes automatically in the event of unintended movement of the portable tank during filling or discharge or fire engulfment. Except for portable tanks having a capacity of not more than 1000 L, it shall be possible to operate this device by remote control.
- 6.7.3.5.5 In addition to filling, discharge and gas pressure equalizing orifices, shells may have openings in which gauges, thermometers and manometers can be fitted. Connections for such instruments shall be made by suitable welded nozzles or pockets and not be screwed connections through the shell.
- 6.7.3.5.6 All portable tanks shall be fitted with manholes or other inspection openings of suitable size to allow for internal inspection and adequate access for maintenance and repair of the interior.
- 6.7.3.5.7 External fittings shall be grouped together so far as reasonably practicable.
- 6.7.3.5.8 Each connection on a portable tank shall be clearly marked to indicate its function.
- 6.7.3.5.9 Each stop-valve or other means of closure shall be designed and constructed to a rated pressure not less than the MAWP of the shell, taking into account the temperatures expected during transport. All stop-valves with a screwed spindle shall close by a clockwise motion of the handwheel. For other stop-valves, the position (open and closed) and direction of closure shall be clearly indicated. All stop-valves shall be designed to prevent unintentional opening.
- 6.7.3.5.10 Piping shall be designed, constructed and installed so as to avoid the risk of damage due to thermal expansion and contraction, mechanical shock and vibration. All piping shall be of suitable metallic material. Welded pipe joints shall be used wherever possible.
- 6.7.3.5.11 Joints in copper tubing shall be brazed or have an equally strong metal union. The melting point of brazing materials shall be no lower than 525°C. The joints shall not decrease the strength of tubing, as may happen when cutting threads.
- 6.7.3.5.12 The burst pressure of all piping and pipe fittings shall be not less than the highest of four times the MAWP of the shell or four times the pressure to which it may be subjected in service by the action of a pump or other device (except pressure relief devices).
- 6.7.3.5.13 Ductile metals shall be used in the construction of valves and accessories.
- 6.7.3.6 **Bottom openings**
- 6.7.3.6.1 Certain non-refrigerated liquefied gases shall not be transported in portable tanks with bottom openings when portable tank instruction T50 in 4.2.5.2.6 indicates that bottom openings are not allowed. There shall be no openings below the liquid level of the shell when it is filled to its maximum permissible filling limit.
- 6.7.3.7 **Pressure relief devices**
- 6.7.3.7.1 Portable tanks shall be provided with one or more spring-loaded pressure relief devices. The pressure relief devices shall open automatically at a pressure not less than the MAWP and be fully open at a pressure equal to 110% of the MAWP. These devices shall, after discharge, close at a pressure not lower than 10% below the pressure at which discharge starts and shall remain closed at all lower pressures. The pressure relief devices shall be of a type that will resist dynamic forces, including liquid surge. Frangible discs not in series with a spring-loaded pressure relief device are not permitted.
- 6.7.3.7.2 Pressure relief devices shall be designed to prevent the entry of foreign matter, the leakage of gas and the development of any dangerous excess pressure.
- 6.7.3.7.3 Portable tanks intended for the transport of certain non-refrigerated liquefied gases identified in portable tank instruction T50 in 4.2.5.2.6 shall have a pressure relief device approved by the competent authority. Unless a portable tank in dedicated service is fitted with an approved relief device constructed of materials compatible with the load, such device shall comprise a frangible disc preceding a spring-loaded device. The space between the frangible disc and the device shall be provided with a pressure gauge or a suitable tell-tale indicator. This arrangement permits the detection of disc rupture, pinholing or leakage which could cause a malfunction of the pressure relief device. The frangible discs shall rupture at a nominal pressure 10% above the start-to-discharge pressure of the relief device.
- 6.7.3.7.4 In the case of multi-purpose portable tanks, the pressure relief devices shall open at a pressure indicated in 6.7.3.7.1 for the gas having the highest maximum allowable pressure of the gases allowed to be transported in the portable tank.

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## 6.7.3.8 Capacity of relief devices

6.7.3.8.1 The combined delivery capacity of the relief devices shall be sufficient that, in the event of total fire engulfment, the pressure (including accumulation) inside the shell does not exceed 120% of the MAWP. Spring-loaded relief devices shall be used to achieve the full relief capacity prescribed. In the case of multi-purpose tanks, the combined delivery capacity of the pressure relief devices shall be taken for the gas which requires the highest delivery capacity of the gases allowed to be transported in portable tanks.

6.7.3.8.1.1 To determine the total required capacity of the relief devices, which shall be regarded as being the sum of the individual capacities of the several devices, the following formula shall be used:

$$Q = 12.4 \frac{FA^{0.82}}{LC} \sqrt{\frac{ZT}{M}}$$

where:

$Q$  = minimum required rate of discharge in cubic metres of air per second ( $\text{m}^3/\text{s}$ ) at standard conditions: 1 bar and  $0^\circ\text{C}$  (273 K);

$F$  = a coefficient with the following value:

for uninsulated shells,  $F = 1$

for insulated shells,  $F = U(649 - t)/13.6$  but in no case is less than 0.25

where:

$\Delta U$  = heat transfer coefficient of the insulation, in  $\text{kW}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$ , at  $38^\circ\text{C}$ ;

$t$  = actual temperature of the non-refrigerated liquefied gas during filling (in  $^\circ\text{C}$ ) (when this temperature is unknown, let  $t = 15^\circ\text{C}$ );

The value of  $F$  given above for insulated shells may be taken provided that the insulation is in conformance with 6.7.3.8.1.2;

$A$  = total external surface area of shell in square metres;

$Z$  = the gas compressibility factor in the accumulating condition (when this factor is unknown, let  $Z$  equal 1.0);

$T$  = absolute temperature in kelvin ( $^\circ\text{C} + 273$ ) above the pressure relief devices in the accumulating condition;

$L$  = the latent heat of vaporization of the liquid, in  $\text{kJ/kg}$ , in the accumulating condition;

$M$  = molecular mass of the discharged gas;

$C$  = a constant which is derived from one of the following formulae as a function of the ratio  $k$  of specific heats:

$$k = \frac{C_p}{C_v}$$

where:

$C_p$  = specific heat at constant pressure; and

$C_v$  = specific heat at constant volume.

When  $k > 1$ :

$$C = \sqrt{k \left( \frac{2}{k+1} \right)^{\frac{k+1}{k-1}}}$$

When  $k = 1$  or  $k$  is unknown:

$$C = \frac{1}{\sqrt{e}} = 0.607$$

where  $e$  is the mathematical constant 2.7183.

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C may also be taken from the following table:

| k    | C     | k    | C     | k    | C     |
|------|-------|------|-------|------|-------|
| 1.00 | 0.607 | 1.26 | 0.660 | 1.52 | 0.704 |
| 1.02 | 0.611 | 1.28 | 0.664 | 1.54 | 0.707 |
| 1.04 | 0.615 | 1.30 | 0.667 | 1.56 | 0.71  |
| 1.06 | 0.620 | 1.32 | 0.671 | 1.58 | 0.713 |
| 1.08 | 0.624 | 1.34 | 0.674 | 1.60 | 0.716 |
| 1.10 | 0.628 | 1.36 | 0.678 | 1.62 | 0.719 |
| 1.12 | 0.633 | 1.38 | 0.681 | 1.64 | 0.722 |
| 1.14 | 0.637 | 1.40 | 0.685 | 1.66 | 0.725 |
| 1.16 | 0.641 | 1.42 | 0.688 | 1.68 | 0.728 |
| 1.18 | 0.645 | 1.44 | 0.691 | 1.70 | 0.731 |
| 1.20 | 0.649 | 1.46 | 0.695 | 2.0  | 0.77  |
| 1.22 | 0.652 | 1.48 | 0.698 | 2.2  | 0.793 |
| 1.24 | 0.656 | 1.50 | 0.701 |      |       |

**Note:** This formula applies only to non-refrigerated liquefied gases which have critical temperatures well above the temperature at the accumulating condition. For gases which have critical temperatures near or below the temperature at the accumulating condition, the calculation of the pressure-relief device delivery capacity shall consider further thermodynamic properties of the gas (see, e.g. CGA S-1.2-2003 Pressure Relief Device Standards – Part 2 – Cargo and Portable Tanks for Compressed Gases).

**6.7.3.8.1.2** Insulation systems, used for the purpose of reducing the venting capacity, shall be approved by the competent authority or its authorized body. In all cases, insulation systems approved for this purpose shall:

- .1 remain effective at all temperatures up to 649°C; and
- .2 be jacketed with a material having a melting point of 700°C or greater.

#### 6.7.3.9 Marking of pressure relief devices

**6.7.3.9.1** Every pressure relief device shall be clearly and permanently marked with the following:

- .1 the pressure (in bar or kPa) at which it is set to discharge;
- .2 the allowable tolerance at the discharge pressure, for spring-loaded devices;
- .3 the reference temperature corresponding to the rated pressure, for frangible discs;
- .4 the rated flow capacity of the device in standard cubic metres of air per second (m<sup>3</sup>/s); and
- .5 the cross sectional flow areas of the spring-loaded pressure relief devices and frangible discs in mm<sup>2</sup>.

When practicable, the following information shall also be shown:

- .6 the manufacturer's name and relevant catalogue number.

**6.7.3.9.2** The rated flow capacity marked on the pressure relief devices shall be determined according to ISO 4126-1:2004 and ISO 4126-7:2004.

#### 6.7.3.10 Connections to pressure relief devices

**6.7.3.10.1** Connections to pressure relief devices shall be of sufficient size to enable the required discharge to pass unrestricted to the safety device. No stop-valve shall be installed between the shell and the pressure relief devices except when duplicate devices are provided for maintenance or other reasons and the stop-valves serving the devices actually in use are locked open or the stop-valves are interlocked so that at least one of the duplicate devices is always operable and capable of meeting the provisions of 6.7.3.8. There shall be no obstruction in an opening leading to a vent or pressure relief device which might restrict or cut off the flow from the shell to that device. Vents from the pressure relief devices, when used, shall deliver the relieved vapour or liquid to the atmosphere in conditions of minimum back-pressure on the relieving device.

#### 6.7.3.11 Siting of pressure relief devices

**6.7.3.11.1** Each pressure relief device inlet shall be situated on top of the shell in a position as near the longitudinal and transverse centre of the shell as reasonably practicable. All pressure relief device inlets shall, under maximum filling conditions, be situated in the vapour space of the shell and the devices shall be so arranged as to ensure that the escaping vapour is discharged unrestrictedly. For flammable non-refrigerated liquefied gases, the escaping vapour shall be directed away from the shell in such a manner that it cannot impinge upon the shell. Protective devices which deflect the flow of vapour are permissible provided the required relief-device capacity is not reduced.

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- 6.7.3.11.2** Arrangements shall be made to prevent access to the pressure relief devices by unauthorized persons and to protect the devices from damage caused by the portable tank overturning.
- 6.7.3.12 Gauging devices**
- 6.7.3.12.1** Unless a portable tank is intended to be filled by mass, it shall be equipped with one or more gauging devices. Glass level-gauges and gauges made of other fragile material, which are in direct communication with the contents of the shell, shall not be used.
- 6.7.3.13 Portable tank supports, frameworks, lifting and tie-down attachments**
- 6.7.3.13.1** Portable tanks shall be designed and constructed with a support structure to provide a secure base during transport. The forces specified in 6.7.3.2.9 and the safety factor specified in 6.7.3.2.10 shall be considered in this aspect of the design. Skids, frameworks, cradles or other similar structures are acceptable.
- 6.7.3.13.2** The combined stresses caused by portable tank mountings (such as cradles, frameworks, etc.) and portable tank lifting and tie-down attachments shall not cause excessive stress in any portion of the shell. Permanent lifting and tie-down attachments shall be fitted to all portable tanks. Preferably they shall be fitted to the portable tank supports but may be secured to reinforcing plates located on the shell at the points of support.
- 6.7.3.13.3** In the design of supports and frameworks, the effects of environmental corrosion shall be taken into account.
- 6.7.3.13.4** Forklift pockets shall be capable of being closed off. The means of closing forklift pockets shall be a permanent part of the framework or permanently attached to the framework. Single-compartment portable tanks with a length less than 3.65 m need not have closed-off forklift pockets provided that:
- .1 the shell and all the fittings are well protected from being hit by the forklift blades; and
  - .2 the distance between the centres of the forklift pockets is at least half of the maximum length of the portable tank.
- 6.7.3.13.5** When portable tanks are not protected during transport, according to 4.2.2.3, the shells and service equipment shall be protected against damage to the shell and service equipment resulting from lateral or longitudinal impact or overturning. External fittings shall be protected so as to preclude the release of the shell contents upon impact or overturning of the portable tank on its fittings. Examples of protection include:
- .1 protection against lateral impact, which may consist of longitudinal bars protecting the shell on both sides at the level of the median line;
  - .2 protection of the portable tank against overturning, which may consist of reinforcement rings or bars fixed across the frame;
  - .3 protection against rear impact, which may consist of a bumper or frame;
  - .4 protection of the shell against damage from impact or overturning by use of an ISO frame in accordance with ISO 1496-3:1995.
- 6.7.3.14 Design approval**
- 6.7.3.14.1** The competent authority or its authorized body shall issue a design approval certificate for any new design of a portable tank. This certificate shall attest that the portable tank has been surveyed by that authority, is suitable for its intended purpose and meets the provisions of this chapter and, when appropriate, the provisions for gases provided in portable tank instruction T50 in 4.2.5.2.6. When a series of portable tanks are manufactured without change in the design, the certificate shall be valid for the entire series. The certificate shall refer to the prototype test report, the gases allowed to be transported, the materials of construction of the shell and an approval number. The approval number shall consist of the distinguishing sign or mark of the State in whose territory the approval was granted, indicated by the distinguishing sign used on vehicles in international road traffic\* and a registration number. Any alternative arrangements according to 6.7.1.2 shall be indicated on the certificate. A design approval may serve for the approval of smaller portable tanks made of materials of the same kind and thickness, by the same fabrication techniques and with identical supports, equivalent closures and other appurtenances.
- 6.7.3.14.2** The prototype test report for the design approval shall include at least the following:
- .1 the results of the applicable framework test specified in ISO 1496-3:1995;
  - .2 the results of the initial inspection and test in 6.7.3.15.3; and
  - .3 the results of the impact test in 6.7.3.15.1, when applicable.
- 6.7.3.15 Inspection and testing**
- 6.7.3.15.1** Portable tanks meeting the definition of *container* in the *International Convention for Safe Containers, 1972* (CSC Convention), as amended, shall not be used unless they are successfully qualified by subjecting a

\* Distinguishing sign of the State of registration used on motor vehicles and trailers in international road traffic, e.g. in accordance with the Geneva Convention on Road Traffic of 1949 or the Vienna Convention on Road Traffic of 1968.

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representative prototype of each design to the Dynamic, Longitudinal Impact Test prescribed in the *Manual of Tests and Criteria*, part IV, section 41. This provision only applies to portable tanks which are constructed according to a design approval certificate which has been issued on or after 1 January 2008.

- 6.7.3.15.2** The shell and items of equipment of each portable tank shall be inspected and tested before being put into service for the first time (initial inspection and test) and thereafter at not more than five-year intervals (5-year periodic inspection and test) with an intermediate periodic inspection and test (2.5-year periodic inspection and test) midway between the 5-year periodic inspections and tests. The 2.5-year periodic inspection and test may be performed within 3 months of the specified date. An exceptional inspection and test shall be performed regardless of the last periodic inspection and test when necessary according to 6.7.3.15.7.
- 6.7.3.15.3** The initial inspection and test of a portable tank shall include a check of the design characteristics, an internal and external examination of the portable tank and its fittings with due regard to the non-refrigerated liquefied gases to be transported, and a pressure test referring to the test pressures according to 6.7.3.3.2. The pressure test may be performed as a hydraulic test or by using another liquid or gas with the agreement of the competent authority or its authorized body. Before the portable tank is placed into service, a leakproofness test and a test of the satisfactory operation of all service equipment shall also be performed. When the shell and its fittings have been pressure-tested separately, they shall be subjected together after assembly to a leakproofness test. All welds subject to full stress level in the shell shall be inspected during the initial test by radiographic, ultrasonic, or another suitable non-destructive test method. This does not apply to the jacket.
- 6.7.3.15.4** The 5-year periodic inspection and test shall include an internal and external examination and, as a general rule, a hydraulic pressure test. Sheathing, thermal insulation and the like shall be removed only to the extent required for reliable appraisal of the condition of the portable tank. When the shell and equipment have been pressure-tested separately, they shall be subjected together after assembly to a leakproofness test.
- 6.7.3.15.5** The intermediate 2.5-year periodic inspection and test shall at least include an internal and external examination of the portable tank and its fittings with due regard to the non-refrigerated liquefied gases intended to be transported, a leakproofness test and a test of the satisfactory operation of all service equipment. Sheathing, thermal insulation and the like shall be removed only to the extent required for reliable appraisal of the condition of the portable tank. For portable tanks intended for the transport of a single non-refrigerated liquefied gas, the 2.5-year internal examination may be waived or substituted by other test methods or inspection procedures specified by the competent authority or its authorized body.
- 6.7.3.15.6** A portable tank may not be filled and offered for transport after the date of expiry of the last 5-year or 2.5-year periodic inspection and test as required by 6.7.3.15.2. However, a portable tank filled prior to the date of expiry of the last periodic inspection and test may be transported for a period not to exceed three months beyond the date of expiry of the last periodic test or inspection. In addition, a portable tank may be transported after the date of expiry of the last periodic test and inspection:
- .1 after emptying but before cleaning, for purposes of performing the next required test or inspection prior to refilling; and
  - .2 unless otherwise approved by the competent authority, for a period not to exceed six months beyond the date of expiry of the last periodic test or inspection, in order to allow the return of dangerous goods for proper disposal or recycling. Reference to this exemption shall be mentioned in the transport document.
- Except as provided for in this paragraph, portable tanks which have missed the timeframe for their scheduled 5-year or 2.5-year periodic inspection and test may only be filled and offered for transport if a new 5-year periodic inspection and test is performed according to 6.7.3.15.4.
- 6.7.3.15.7** The exceptional inspection and test is necessary when the portable tank shows evidence of damaged or corroded areas, or leakage, or other conditions that indicate a deficiency that could affect the integrity of the portable tank. The extent of the exceptional inspection and test shall depend on the amount of damage or deterioration of the portable tank. It shall include at least the 2.5-year inspection and test according to 6.7.3.15.5.
- 6.7.3.15.8** The internal and external examinations shall ensure that:
- .1 the shell is inspected for pitting, corrosion, or abrasions, dents, distortions, defects in welds or any other conditions, including leakage, that might render the portable tank unsafe for transport. The wall thickness shall be verified by appropriate measurement if this inspection indicates a reduction of wall thickness;
  - .2 the piping, valves, and gaskets are inspected for corroded areas, defects, or any other conditions, including leakage, that might render the portable tank unsafe for filling, discharge or transport;
  - .3 devices for tightening manhole covers are operative and there is no leakage at manhole covers or gaskets;
  - .4 missing or loose bolts or nuts on any flanged connection or blank flange are replaced or tightened;
  - .5 all emergency devices and valves are free from corrosion, distortion and any damage or defect that could prevent their normal operation. Remote closure devices and self-closing stop-valves shall be operated to demonstrate proper operation;



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- .6 required marks on the portable tank are legible and in accordance with the applicable provisions; and
- .7 the framework, the supports and the arrangements for lifting the portable tank are in satisfactory condition.

**6.7.3.15.9** The inspections and tests in 6.7.3.15.1, 6.7.3.15.3, 6.7.3.15.4, 6.7.3.15.5 and 6.7.3.15.7 shall be performed or witnessed by an expert approved by the competent authority or its authorized body. When the pressure test is a part of the inspection and test, the test pressure shall be the one indicated on the data plate of the portable tank. While under pressure, the portable tank shall be inspected for any leaks in the shell, piping or equipment.

**6.7.3.15.10** In all cases when cutting, burning or welding operations on the shell have been effected, that work shall be to the approval of the competent authority or its authorized body, taking into account the pressure-vessel code used for the construction of the shell. A pressure test to the original test pressure shall be performed after the work is completed.

**6.7.3.15.11** When evidence of any unsafe condition is discovered, the portable tank shall not be returned to service until it has been corrected and the pressure test is repeated and passed.

**6.7.3.16 Marking**

**6.7.3.16.1** Every portable tank shall be fitted with a corrosion-resistant metal plate permanently attached to the portable tank in a conspicuous place readily accessible for inspection. When for reasons of portable tank arrangements the plate cannot be permanently attached to the shell, the shell shall be marked with at least the information required by the pressure-vessel code. As a minimum, at least the following information shall be marked on the plate by stamping or by any other similar method:

- (a) owner information
  - (i) owner's registration number;
- (b) manufacturing information
  - (i) country of manufacture;
  - (ii) year of manufacture;
  - (iii) manufacturer's name or mark;
  - (iv) manufacturer's serial number;
- (c) approval information
  - (i) the United Nations packaging symbol:



This symbol shall not be used for any purpose other than certifying that a packaging, a flexible bulk container, a portable tank or a MEGC complies with the relevant requirements in chapters 6.1, 6.2, 6.3, 6.5, 6.6, 6.7 or 6.9;

- (ii) approval country;
- (iii) authorized body for the design approval;
- (iv) design approval number;
- (v) letters "AA", if the design was approved under alternative arrangements (see 6.7.1.2);
- (vi) pressure-vessel code to which the shell is designed;
- (d) pressures
  - (i) MAWP (in bar gauge or kPa gauge);\*
  - (ii) test pressure (in bar gauge or kPa gauge);\*
  - (iii) initial pressure test date (month and year);
  - (iv) identification mark of the initial pressure test witness;
  - (v) external design pressure<sup>†</sup> (in bar gauge or kPa gauge);\*
- (e) temperatures
  - (i) design temperature range (in °C);\*
  - (ii) design reference temperature (in °C);\*
- (f) materials
  - (i) shell material(s) and material standard reference(s);
  - (ii) equivalent thickness in reference steel (in mm);\*

\* The unit used shall be indicated.

<sup>†</sup> See 6.7.3.2.8.

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- (g) capacity
  - (i) tank water capacity at 20°C (in litres);\*
- (h) periodic inspections and tests
  - (i) type of the most recent periodic test (2.5-year, 5-year or exceptional);
  - (ii) date of the most recent periodic test (month and year);
  - (iii) test pressure (in bar gauge or kPa gauge)\* of the most recent periodic test (if applicable);
  - (iv) identification mark of the authorized body who performed or witnessed the most recent test.

Figure 6.7.3.16.1 – Example of a plate for marking

|                                                                                    |                                     |                                              |            |                |           |                                              |                      |
|------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------------|------------|----------------|-----------|----------------------------------------------|----------------------|
| Owner's registration number                                                        |                                     |                                              |            |                |           |                                              |                      |
| <b>MANUFACTURING INFORMATION</b>                                                   |                                     |                                              |            |                |           |                                              |                      |
| Country of manufacture                                                             |                                     |                                              |            |                |           |                                              |                      |
| Year of manufacture                                                                |                                     |                                              |            |                |           |                                              |                      |
| Manufacturer                                                                       |                                     |                                              |            |                |           |                                              |                      |
| Manufacturer's serial number                                                       |                                     |                                              |            |                |           |                                              |                      |
| <b>APPROVAL INFORMATION</b>                                                        |                                     |                                              |            |                |           |                                              |                      |
|  | Approval country                    |                                              |            |                |           |                                              |                      |
|                                                                                    | Authorized body for design approval |                                              |            |                |           |                                              |                      |
|                                                                                    | Design approval number              |                                              |            |                |           |                                              | “AA” (if applicable) |
| Shell design code (pressure-vessel code)                                           |                                     |                                              |            |                |           |                                              |                      |
| <b>PRESSURES</b>                                                                   |                                     |                                              |            |                |           |                                              |                      |
| MAWP                                                                               |                                     |                                              |            | bar or kPa     |           |                                              |                      |
| Test pressure                                                                      |                                     |                                              |            | bar or kPa     |           |                                              |                      |
| Initial pressure test date:                                                        |                                     | (mm/yyyy)                                    |            | Witness stamp: |           |                                              |                      |
| External design pressure                                                           |                                     |                                              |            | bar or kPa     |           |                                              |                      |
| <b>TEMPERATURES</b>                                                                |                                     |                                              |            |                |           |                                              |                      |
| Design temperature range                                                           |                                     |                                              |            | °C to °C       |           |                                              |                      |
| Design reference temperature                                                       |                                     |                                              |            | °C             |           |                                              |                      |
| <b>MATERIALS</b>                                                                   |                                     |                                              |            |                |           |                                              |                      |
| Shell material(s) and material standard reference(s)                               |                                     |                                              |            |                |           |                                              |                      |
| Equivalent thickness in reference steel                                            |                                     |                                              |            | mm             |           |                                              |                      |
| <b>CAPACITY</b>                                                                    |                                     |                                              |            |                |           |                                              |                      |
| Tank water capacity at 20°C                                                        |                                     |                                              |            | litres         |           |                                              |                      |
| <b>PERIODIC INSPECTIONS/TESTS</b>                                                  |                                     |                                              |            |                |           |                                              |                      |
| Test type                                                                          | Test date                           | Witness stamp and test pressure <sup>a</sup> |            | Test type      | Test date | Witness stamp and test pressure <sup>a</sup> |                      |
|                                                                                    | (mm/yyyy)                           |                                              | bar or kPa |                | (mm/yyyy) |                                              | bar or kPa           |
|                                                                                    |                                     |                                              |            |                |           |                                              |                      |
|                                                                                    |                                     |                                              |            |                |           |                                              |                      |
|                                                                                    |                                     |                                              |            |                |           |                                              |                      |

<sup>a</sup> Test pressure if applicable.

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- 6.7.3.16.2** The following information shall be durably marked either on the portable tank itself or on a metal plate firmly secured to the portable tank:
- Name of the operator
  - Name of non-refrigerated liquefied gas(es) permitted for transport
  - Maximum permissible load mass for each non-refrigerated liquefied gas permitted . . . . . kg
  - Maximum permissible gross mass (MPGM) . . . . . kg
  - Unladen (tare) mass . . . . . kg
  - Portable tank instruction in accordance with 4.2.5.2.6.
- 6.7.3.16.3** If a portable tank is designed and approved for handling in open seas, the words “OFFSHORE PORTABLE TANK” shall be marked on the identification plate.

## **6.7.4 Provisions for the design, construction, inspection and testing of portable tanks intended for the transport of refrigerated liquefied gases of class 2**

### **6.7.4.1 Definitions**

For the purposes of this section:

*Holding time* means the time that will elapse from the establishment of the initial filling condition until the pressure has risen due to heat influx to the lowest set pressure of the pressure-limiting device(s);

*Jacket* means the outer insulation cover or cladding, which may be part of the insulation system;

*Leakproofness test* means a test, using gas, subjecting the shell and its service equipment to an effective internal pressure not less than 90% of the MAWP;

*Maximum allowable working pressure (MAWP)* means the maximum effective gauge pressure permissible at the top of the shell of a filled portable tank in its operating position, including the highest effective pressure during filling and discharge;

*Maximum permissible gross mass (MPGM)* means the sum of the tare mass of the portable tank and the heaviest load authorized for transport;

*Minimum design temperature* means the temperature which is used for the design and construction of the shell, not higher than the lowest (coldest) temperature (service temperature) of the contents during normal conditions of filling, discharge and transport;

*Portable tank* means a thermally insulated multimodal tank having a capacity of more than 450 L fitted with service equipment and structural equipment necessary for the transport of refrigerated liquefied gases. The portable tank shall be capable of being filled and discharged without the removal of its structural equipment. It shall possess stabilizing members external to the tank, and shall be capable of being lifted when full. It shall be designed primarily to be loaded onto a transport vehicle or ship and shall be equipped with skids, mountings or accessories to facilitate mechanical handling. Road tank-vehicles, rail tank-wagons, non-metallic tanks, intermediate bulk containers (IBCs), gas cylinders and large receptacles are not considered to fall within the definition for portable tanks;

*Reference steel* means a steel with a tensile strength of 370 N/mm<sup>2</sup> and an elongation at fracture of 27%;

*Service equipment* means measuring instruments and filling, discharge, venting, safety, pressurizing, cooling and thermal insulation devices;

*Shell* means the part of the portable tank which retains the refrigerated liquefied gas intended for transport, including openings and their closures, but does not include service equipment or external structural equipment;

*Structural equipment* means the reinforcing, fastening, protective and stabilizing members external to the shell;

*Tank* means a construction which normally consists of either:

- (a) a jacket and one or more inner shells where the space between the shell(s) and the jacket is exhausted of air (vacuum insulation) and may incorporate a thermal insulation system; or
- (b) a jacket and an inner shell with an intermediate layer of solid thermally insulating material (such as solid foam);

*Test pressure* means the maximum gauge pressure at the top of the shell during the pressure test.

### **6.7.4.2 General design and construction provisions**

- 6.7.4.2.1** Shells shall be designed and constructed in accordance with the provisions of a pressure-vessel code recognized by the competent authority. Shells and jackets shall be made of metallic materials suitable for forming. Jackets shall be made of steel. Non-metallic materials may be used for the attachments and supports between the shell and jacket, provided their material properties at the minimum design temperature are proven to be sufficient. The materials shall, in principle, conform to national or international material

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standards. For welded shells and jackets, only materials whose weldability has been fully demonstrated shall be used. Welds shall be skilfully made and afford complete safety. When the manufacturing process or the materials make it necessary, the shell shall be suitably heat-treated to guarantee adequate toughness in the weld and in the heat-affected zones. In choosing the material, the minimum design temperature shall be taken into account with respect to risk of brittle fracture, to hydrogen embrittlement, to stress corrosion cracking and to resistance to impact. When fine-grain steel is used, the guaranteed value of the yield strength shall be not more than 460 N/mm<sup>2</sup> and the guaranteed value of the upper limit of the tensile strength shall be not more than 725 N/mm<sup>2</sup>, in accordance with the material specifications. Portable tank materials shall be suitable for the external environment in which they may be transported.

- 6.7.4.2.2 Any part of a portable tank, including fittings, gaskets and pipe-work, which can be expected normally to come into contact with the refrigerated liquefied gas transported shall be compatible with that refrigerated liquefied gas.
- 6.7.4.2.3 Contact between dissimilar metals which could result in damage by galvanic action shall be avoided.
- 6.7.4.2.4 The thermal insulation system shall include a complete covering of the shell(s) with effective insulating materials. External insulation shall be protected by a jacket so as to prevent the ingress of moisture and other damage under normal transport conditions.
- 6.7.4.2.5 When a jacket is so closed as to be gas-tight, a device shall be provided to prevent any dangerous pressure from developing in the insulation space.
- 6.7.4.2.6 Portable tanks intended for the transport of refrigerated liquefied gases having a boiling point below –182°C at atmospheric pressure shall not include materials which may react with oxygen or oxygen-enriched atmospheres in a dangerous manner when located in parts of the thermal insulation when there is a risk of contact with oxygen or with oxygen-enriched fluid.
- 6.7.4.2.7 Insulating materials shall not deteriorate unduly in service.
- 6.7.4.2.8 A reference holding time shall be determined for each refrigerated liquefied gas intended for transport in a portable tank.
  - 6.7.4.2.8.1 The reference holding time shall be determined by a method recognized by the competent authority on the basis of the following:
    - .1 the effectiveness of the insulation system, determined in accordance with 6.7.4.2.8.2;
    - .2 the lowest set pressure of the pressure-limiting device(s);
    - .3 the initial filling conditions;
    - .4 an assumed ambient temperature of 30°C;
    - .5 the physical properties of the individual refrigerated liquefied gas intended to be transported.
  - 6.7.4.2.8.2 The effectiveness of the insulation system (heat influx in watts) shall be determined by type testing the portable tank in accordance with a procedure recognized by the competent authority. This test shall consist of either:
    - .1 a constant-pressure test (for example at atmospheric pressure), when the loss of refrigerated liquefied gas is measured over a period of time; or
    - .2 a closed-system test, when the rise in pressure in the shell is measured over a period of time.

When performing the constant-pressure test, variations in atmospheric pressure shall be taken into account. When performing either test, corrections shall be made for any variation of the ambient temperature from the assumed ambient temperature reference value of 30°C.

**Note:** For the determination of the actual holding time before each journey, see 4.2.3.7.
- 6.7.4.2.9 The jacket of a vacuum-insulated double-wall tank shall have either an external design pressure not less than 100 kPa (1 bar) gauge pressure calculated in accordance with a recognized technical code or a calculated critical collapsing pressure of not less than 200 kPa (2 bar) gauge pressure. Internal and external reinforcements may be included in calculating the ability of the jacket to resist the external pressure.
- 6.7.4.2.10 Portable tanks shall be designed and constructed with supports to provide a secure base during transport and with suitable lifting and tie-down attachments.
- 6.7.4.2.11 Portable tanks shall be designed to withstand, without loss of contents, at least the internal pressure due to the contents and the static, dynamic and thermal loads during normal conditions of handling and transport. The design shall demonstrate that the effects of fatigue, caused by repeated application of these loads through the expected life of the portable tank, have been taken into account.
  - 6.7.4.2.11.1 For tanks that are intended for use as offshore tank-containers, the dynamic stresses imposed by handling in open seas shall be taken into account.

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- 6.7.4.2.12** Portable tanks and their fastenings under the maximum permissible load shall be capable of absorbing the following separately applied static forces:
- .1 in the direction of travel: twice the MPGM multiplied by the acceleration due to gravity ( $g$ );\*
  - .2 horizontally at right angles to the direction of travel: the MPGM (when the direction of travel is not clearly determined, the forces shall be equal to twice the MPGM) multiplied by the acceleration due to gravity ( $g$ );\*
  - .3 vertically upwards: the MPGM multiplied by the acceleration due to gravity ( $g$ );\* and
  - .4 vertically downwards: twice the MPGM (total loading including the effect of gravity) multiplied by the acceleration due to gravity ( $g$ );\*
- 6.7.4.2.13** Under each of the forces in 6.7.4.2.12, the safety factor to be observed shall be as follows:
- .1 for materials having a clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed yield strength; or
  - .2 for materials with no clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed 0.2% proof strength or, for austenitic steels, the 1% proof strength.
- 6.7.4.2.14** The values of yield strength or proof strength shall be the values according to national or international material standards. When austenitic steels are used, the specified minimum values according to the material standards may be increased by up to 15% when these greater values are attested in the material inspection certificate. When no material standard exists for the metal in question, or when non-metallic materials are used, the values of yield strength or proof strength shall be approved by the competent authority.
- 6.7.4.2.15** Portable tanks intended for the transport of flammable refrigerated liquefied gases shall be capable of being electrically earthed.
- 6.7.4.3 Design criteria**
- 6.7.4.3.1** Shells shall be of a circular cross-section.
- 6.7.4.3.2** Shells shall be designed and constructed to withstand a test pressure not less than 1.3 times the MAWP. For shells with vacuum insulation, the test pressure shall not be less than 1.3 times the sum of the MAWP and 100 kPa (1 bar). In no case shall the test pressure be less than 300 kPa (3 bar) gauge pressure. Attention is drawn to the minimum shell thickness provisions, specified in 6.7.4.4.2 to 6.7.4.4.7.
- 6.7.4.3.3** For metals exhibiting a clearly defined yield point or characterized by a guaranteed proof strength (0.2% proof strength, generally, or 1% proof strength for austenitic steels), the primary membrane stress  $\sigma$  (sigma) in the shell shall not exceed  $0.75R_e$  or  $0.50R_m$ , whichever is lower, at the test pressure, where:
- $R_e$  = yield strength in N/mm<sup>2</sup>, or 0.2% proof strength or, for austenitic steels, 1% proof strength;
- $R_m$  = minimum tensile strength in N/mm<sup>2</sup>.
- 6.7.4.3.3.1** The values of  $R_e$  and  $R_m$  to be used shall be the specified minimum values according to national or international material standards. When austenitic steels are used, the specified minimum values for  $R_e$  and  $R_m$  according to the material standards may be increased by up to 15% when greater values are attested in the material inspection certificate. When no material standard exists for the metal in question, the values of  $R_e$  and  $R_m$  used shall be approved by the competent authority or its authorized body.
- 6.7.4.3.3.2** Steels which have a  $R_e/R_m$  ratio of more than 0.85 are not allowed for the construction of welded shells. The values of  $R_e$  and  $R_m$  to be used in determining this ratio shall be the values specified in the material inspection certificate.
- 6.7.4.3.3.3** Steels used in the construction of shells shall have an elongation at fracture, in %, of not less than  $10,000/R_m$  with an absolute minimum of 16% for fine-grain steels and 20% for other steels. Aluminium and aluminium alloys used in the construction of shells shall have an elongation at fracture, in %, of not less than  $10,000/6R_m$  with an absolute minimum of 12%.
- 6.7.4.3.3.4** For the purpose of determining actual values for materials, it shall be noted that for sheet metal, the axis of the tensile test specimen shall be at right angles (transversely) to the direction of rolling. The permanent elongation at fracture shall be measured on test specimens of rectangular cross-section in accordance with ISO 6892:1998 using a 50 mm gauge length.
- 6.7.4.4 Minimum shell thickness**
- 6.7.4.4.1** The minimum shell thickness shall be the greater thickness based on:
- .1 the minimum thickness determined in accordance with the provisions in 6.7.4.4.2 to 6.7.4.4.7; and

\* For calculation purposes,  $g = 9.81 \text{ m/s}^2$ .

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.2 the minimum thickness determined in accordance with the recognized pressure-vessel code, including the provisions in 6.7.4.3.

6.7.4.4.2 Shells of not more than 1.80 m in diameter shall be not less than 5 mm thick in the reference steel or of equivalent thickness in the metal to be used. Shells of more than 1.80 m in diameter shall be not less than 6 mm thick in the reference steel or of equivalent thickness in the metal to be used.

6.7.4.4.3 Shells of vacuum-insulated tanks of not more than 1.80 m in diameter shall be not less than 3 mm thick in the reference steel or of equivalent thickness in the metal to be used. Such shells of more than 1.80 m in diameter shall be not less than 4 mm thick in the reference steel or of equivalent thickness in the metal to be used.

6.7.4.4.4 For vacuum-insulated tanks, the aggregate thickness of the jacket and the shell shall correspond to the minimum thickness prescribed in 6.7.4.4.2, the thickness of the shell itself being not less than the minimum thickness prescribed in 6.7.4.4.3.

6.7.4.4.5 Shells shall be not less than 3 mm thick regardless of the material of construction.

6.7.4.4.6 The equivalent thickness of a metal other than the thickness prescribed for the reference steel in 6.7.4.4.2 and 6.7.4.4.3 shall be determined using the following equation:

$$e_1 = \frac{21.4 \times e_0}{\sqrt[3]{R_{m1} \times A_1}}$$

where:

$e_1$  = required equivalent thickness (in mm) of the steel to be used;

$e_0$  = minimum thickness (in mm) of the reference steel specified in 6.7.4.4.2 and 6.7.4.4.3;

$R_{m1}$  = guaranteed minimum tensile strength (in N/mm<sup>2</sup>) of the metal to be used (see 6.7.4.3.3);

$A_1$  = guaranteed minimum elongation at fracture (in %) of the metal to be used according to national or international standards.

6.7.4.4.7 In no case shall the wall thickness be less than that prescribed in 6.7.4.4.1 to 6.7.4.4.5. All parts of the shell shall have a minimum thickness as determined by 6.7.4.4.1 to 6.7.4.4.6. This thickness shall be exclusive of any corrosion allowance.

6.7.4.4.8 There shall be no sudden change of plate thickness at the attachment of the ends (heads) to the cylindrical portion of the shell.

#### 6.7.4.5 Service equipment

6.7.4.5.1 Service equipment shall be so arranged as to be protected against the risk of being wrenched off or damaged during handling and transport. When the connection between the frame and the tank or the jacket and the shell allows relative movement, the equipment shall be so fastened as to permit such movement without risk of damage to working parts. The external discharge fittings (pipe sockets, shut-off devices), the stop-valve and its seating shall be protected against the danger of being wrenched off by external forces (for example, by using shear sections). The filling and discharge devices (including flanges or threaded plugs) and any protective caps shall be capable of being secured against unintended opening.

6.7.4.5.1.1 For offshore tank-containers, where positioning of service equipment and the design and strength of protection for such equipment is concerned, the increased danger of impact damage when handling such tanks in open seas shall be taken into account.

6.7.4.5.2 Each filling and discharge opening in portable tanks used for the transport of flammable refrigerated liquefied gases shall be fitted with at least three mutually independent shut-off devices in series, the first being a stop-valve situated as close as reasonably practicable to the jacket, the second being a stop-valve and the third being a blank flange or equivalent device. The shut-off device closest to the jacket shall be a quick-closing device, which closes automatically in the event of unintended movement of the portable tank during filling or discharge or fire engulfment. This device shall also be possible to operate by remote control.

6.7.4.5.3 Each filling and discharge opening in portable tanks used for the transport of non-flammable refrigerated liquefied gases shall be fitted with at least two mutually independent shut-off devices in series, the first being a stop-valve situated as close as reasonably practicable to the jacket, the second a blank flange or equivalent device.

6.7.4.5.4 For sections of piping which can be closed at both ends and where liquid product can be trapped, a method of automatic pressure relief shall be provided to prevent excess pressure build-up within the piping.

6.7.4.5.5 Vacuum-insulated tanks need not have an opening for inspection.

6.7.4.5.6 External fittings shall be grouped together so far as reasonably practicable.

6.7.4.5.7 Each connection on a portable tank shall be clearly marked to indicate its function.



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- 6.7.4.5.8** Each stop-valve or other means of closure shall be designed and constructed to a rated pressure not less than the MAWP of the shell, taking into account the temperature expected during transport. All stop-valves with a screwed spindle shall be closed by a clockwise motion of the handwheel. In the case of other stop-valves, the position (open and closed) and direction of closure shall be clearly indicated. All stop-valves shall be designed to prevent unintentional opening.
- 6.7.4.5.9** When pressure-building units are used, the liquid and vapour connections to that unit shall be provided with a valve as close to the jacket as reasonably practicable to prevent the loss of contents in case of damage to the pressure-building unit.
- 6.7.4.5.10** Piping shall be designed, constructed and installed so as to avoid the risk of damage due to thermal expansion and contraction, mechanical shock and vibration. All piping shall be of a suitable material. To prevent leakage due to fire, only steel piping and welded joints shall be used between the jacket and the connection to the first closure of any outlet. The method of attaching the closure to this connection shall be to the satisfaction of the competent authority or its authorized body. Elsewhere, pipe joints shall be welded when necessary.
- 6.7.4.5.11** Joints in copper tubing shall be brazed or have an equally strong metal union. The melting point of brazing materials shall be no lower than 525°C. The joints shall not decrease the strength of the tubing, as may happen by cutting of threads.
- 6.7.4.5.12** The materials of construction of valves and accessories shall have satisfactory properties at the lowest operating temperature of the portable tank.
- 6.7.4.5.13** The burst pressure of all piping and pipe fittings shall be not less than the highest of four times the MAWP of the shell or four times the pressure to which it may be subjected in service by the action of a pump or other device (except pressure relief devices).
- 6.7.4.6 Pressure relief devices**
- 6.7.4.6.1** Every shell shall be provided with not less than two independent spring-loaded pressure relief devices. The pressure relief devices shall open automatically at a pressure not less than the MAWP and be fully open at a pressure equal to 110% of the MAWP. These devices shall, after discharge, close at a pressure not lower than 10% below the pressure at which discharge starts and shall remain closed at all lower pressures. The pressure relief devices shall be of the type that will resist dynamic forces, including surge.
- 6.7.4.6.2** Shells for non-flammable refrigerated liquefied gases and hydrogen may in addition have frangible discs in parallel with the spring-loaded devices as specified in 6.7.4.7.2 and 6.7.4.7.3.
- 6.7.4.6.3** Pressure relief devices shall be designed to prevent the entry of foreign matter, the leakage of gas and the development of any dangerous excess pressure.
- 6.7.4.6.4** Pressure relief devices shall be approved by the competent authority or its authorized body.
- 6.7.4.7 Capacity and setting of pressure relief devices**
- 6.7.4.7.1** In the case of the loss of vacuum in a vacuum-insulated tank or of loss of 20% of the insulation of a tank insulated with solid materials, the combined capacity of all pressure relief devices installed shall be sufficient so that the pressure (including accumulation) inside the shell does not exceed 120% of the MAWP.
- 6.7.4.7.2** For non-flammable refrigerated liquefied gases (except oxygen) and hydrogen, this capacity may be achieved by the use of frangible discs in parallel with the required safety relief devices. Frangible discs shall rupture at nominal pressure equal to the test pressure of the shell.
- 6.7.4.7.3** Under the circumstances described in 6.7.4.7.1 and 6.7.4.7.2 together with complete fire engulfment, the combined capacity of all pressure relief devices installed shall be sufficient to limit the pressure in the shell to the test pressure.
- 6.7.4.7.4** The required capacity of the relief devices shall be calculated in accordance with a well-established technical code recognized by the competent authority.\*
- 6.7.4.8 Marking of pressure relief devices**
- 6.7.4.8.1** Every pressure relief device shall be plainly and permanently marked with the following:
- .1 the pressure (in bar or kPa) at which it is set to discharge;
  - .2 the allowable tolerance at the discharge pressure, for spring-loaded devices;
  - .3 the reference temperature corresponding to the rated pressure, for frangible discs;

\* See, e.g. CGA S-1.2-2003 *Pressure Relief Device Standards – Part 2 – Cargo and Portable Tanks for Compressed Gases*.

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- .4 the rated flow capacity of the device in standard cubic metres of air per second (m<sup>3</sup>/s); and
- .5 the cross sectional flow areas of the spring-loaded pressure relief devices and frangible discs in mm<sup>2</sup>.

When practicable, the following information shall also be shown:

- .6 the manufacturer's name and relevant catalogue number.

6.7.4.8.2 The rated flow capacity marked on the pressure relief devices shall be determined according to ISO 4126-1:2004 and ISO 4126-7:2004.

#### 6.7.4.9 Connections to pressure relief devices

6.7.4.9.1 Connections to pressure relief devices shall be of sufficient size to enable the required discharge to pass unrestricted to the safety device. No stop-valve shall be installed between the shell and the pressure relief devices except when duplicate devices are provided for maintenance or other reasons and the stop-valves serving the devices actually in use are locked open or the stop-valves are interlocked so that the provisions of 6.7.4.7 are always fulfilled. There shall be no obstruction in an opening leading to a vent or pressure relief device which might restrict or cut off the flow from the shell to that device. Pipework to vent the vapour or liquid from the outlet of the pressure relief devices, when used, shall deliver the relieved vapour or liquid to the atmosphere in conditions of minimum back-pressure on the relieving device.

#### 6.7.4.10 Siting of pressure relief devices

6.7.4.10.1 Each pressure relief device inlet shall be situated on top of the shell in a position as near the longitudinal and transverse centre of the shell as reasonably practicable. All pressure relief device inlets shall, under maximum filling conditions, be situated in the vapour space of the shell and the devices shall be so arranged as to ensure that the escaping vapour is discharged unrestrictedly. For refrigerated liquefied gases, the escaping vapour shall be directed away from the tank and in such a manner that it cannot impinge upon the tank. Protective devices which deflect the flow of vapour are permissible provided the required relief-device capacity is not reduced.

6.7.4.10.2 Arrangements shall be made to prevent access to the devices by unauthorized persons and to protect the devices from damage caused by the portable tank overturning.

#### 6.7.4.11 Gauging devices

6.7.4.11.1 Unless a portable tank is intended to be filled by mass, it shall be equipped with one or more gauging devices. Glass level-gauges and gauges made of other fragile material, which are in direct communication with the contents of the shell, shall not be used.

6.7.4.11.2 A connection for a vacuum gauge shall be provided in the jacket of a vacuum-insulated portable tank.

#### 6.7.4.12 Portable tank supports, frameworks, lifting and tie-down attachments

6.7.4.12.1 Portable tanks shall be designed and constructed with a support structure to provide a secure base during transport. The forces specified in 6.7.4.2.12 and the safety factor specified in 6.7.4.2.13 shall be considered in this aspect of the design. Skids, frameworks, cradles or other similar structures are acceptable.

6.7.4.12.2 The combined stresses caused by portable tank mountings (such as cradles, frameworks, etc.) and portable tank lifting and tie-down attachments shall not cause excessive stress in any portion of the tank. Permanent lifting and tie-down attachments shall be fitted to all portable tanks. Preferably they shall be fitted to the portable tank supports but may be secured to reinforcing plates located on the tank at the points of support.

6.7.4.12.3 In the design of supports and frameworks, the effects of environmental corrosion shall be taken into account.

6.7.4.12.4 Forklift pockets shall be capable of being closed off. The means of closing forklift pockets shall be a permanent part of the framework or permanently attached to the framework. Single-compartment portable tanks with a length less than 3.65 m need not have closed-off forklift pockets provided that:

- .1 the tank and all the fittings are well protected from being hit by the forklift blades; and
- .2 the distance between the centres of the forklift pockets is at least half of the maximum length of the portable tank.

6.7.4.12.5 When portable tanks are not protected during transport, according to 4.2.3.3, the shells and service equipment shall be protected against damage to the shell and service equipment resulting from lateral or longitudinal impact or overturning. External fittings shall be protected so as to preclude the release of the shell contents upon impact or overturning of the portable tank on its fittings. Examples of protection include:

- .1 protection against lateral impact, which may consist of longitudinal bars protecting the shell on both sides at the level of the median line;

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- .2 protection of the portable tank against overturning, which may consist of reinforcement rings or bars fixed across the frame;
- .3 protection against rear impact, which may consist of a bumper or frame;
- .4 protection of the shell against damage from impact or overturning by use of an ISO frame in accordance with ISO 1496-3:1995;
- .5 protection of the portable tank from impact or overturning by a vacuum insulation jacket.

**6.7.4.13 Design approval**

**6.7.4.13.1** The competent authority or its authorized body shall issue a design approval certificate for any new design of a portable tank. This certificate shall attest that a portable tank has been surveyed by that authority, is suitable for its intended purpose and meets the provisions of this chapter. When a series of portable tanks are manufactured without change in the design, the certificate shall be valid for the entire series. The certificate shall refer to the prototype test report, the refrigerated liquefied gases allowed to be transported, the materials of construction of the shell and jacket and an approval number. The approval number shall consist of the distinguishing sign or mark of the State in whose territory the approval was granted, indicated by the distinguishing sign used on vehicles in international road traffic\* and a registration number. Any alternative arrangements according to 6.7.1.2 shall be indicated on the certificate. A design approval may serve for the approval of smaller portable tanks made of materials of the same kind and thickness, by the same fabrication techniques and with identical supports, equivalent closures and other appurtenances.

**6.7.4.13.2** The prototype test report for the design approval shall include at least the following:

- .1 the results of the applicable framework test specified in ISO 1496-3:1995;
- .2 the results of the initial inspection and test in 6.7.4.14.3; and
- .3 the results of the impact test in 6.7.4.14.1, when applicable.

**6.7.4.14 Inspection and testing**

**6.7.4.14.1** Portable tanks meeting the definition of *container* in the *International Convention for Safe Containers, 1972* (CSC Convention), as amended, shall not be used unless they are successfully qualified by subjecting a representative prototype of each design to the Dynamic, Longitudinal Impact Test prescribed in the *Manual of Tests and Criteria*, part IV, section 41. This provision only applies to portable tanks which are constructed according to a design approval certificate which has been issued on or after 1 January 2008.

**6.7.4.14.2** The tank and items of equipment of each portable tank shall be inspected and tested before being put into service for the first time (initial inspection and test) and thereafter at not more than five-year intervals (5-year periodic inspection and test) with an intermediate periodic inspection and test (2.5-year periodic inspection and test) midway between the 5-year periodic inspections and tests. The 2.5-year periodic inspection and test may be performed within 3 months of the specified date. An exceptional inspection and test shall be performed regardless of the last periodic inspection and test when necessary according to 6.7.4.14.7.

**6.7.4.14.3** The initial inspection and test of a portable tank shall include a check of the design characteristics, an internal and external examination of the portable tank shell and its fittings with due regard to the refrigerated liquefied gases to be transported, and a pressure test referring to the test pressures according to 6.7.4.3.2. The pressure test may be performed as a hydraulic test or by using another liquid or gas, with the agreement of the competent authority or its authorized body. Before the portable tank is placed into service, a leakproofness test and a test of the satisfactory operation of all service equipment shall also be performed. When the shell and its fittings have been pressure-tested separately, they shall be subjected together after assembly to a leakproofness test. All welds subject to full stress level shall be inspected during the initial test by radiographic, ultrasonic, or another suitable non-destructive test method. This does not apply to the jacket.

**6.7.4.14.4** The 5-year and 2.5-year periodic inspections and tests shall include an external examination of the portable tank and its fittings with due regard to the refrigerated liquefied gases transported, a leakproofness test, a test of the satisfactory operation of all service equipment and a vacuum reading, when applicable. In the case of non-vacuum-insulated tanks, the jacket and insulation shall be removed during the 2.5-year and the 5-year periodic inspections and tests, but only to the extent necessary for a reliable appraisal.

**6.7.4.14.5** [Reserved]

**6.7.4.14.6** A portable tank may not be filled and offered for transport after the date of expiry of the last 5-year or 2.5-year periodic inspection and test as required by 6.7.4.14.2. However, a portable tank filled prior to the date of expiry of the last periodic inspection and test may be transported for a period not to exceed three months beyond

\* Distinguishing sign of the State of registration used on motor vehicles and trailers in international road traffic, e.g. in accordance with the Geneva Convention on Road Traffic of 1949 or the Vienna Convention on Road Traffic of 1968.

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the date of expiry of the last periodic test or inspection. In addition, a portable tank may be transported after the date of expiry of the last periodic test and inspection:

- .1 after emptying but before cleaning, for purposes of performing the next required test or inspection prior to refilling; and
- .2 unless otherwise approved by the competent authority, for a period not to exceed six months beyond the date of expiry of the last periodic test or inspection, in order to allow the return of dangerous goods for proper disposal or recycling. Reference to this exemption shall be mentioned in the transport document.

Except as provided for in this paragraph, portable tanks which have missed the timeframe for their scheduled 5-year or 2.5-year periodic inspection and test may only be filled and offered for transport if a new 5-year periodic inspection and test is performed according to 6.7.4.14.4.

- 6.7.4.14.7** The exceptional inspection and test is necessary when the portable tank shows evidence of damaged or corroded areas, leakage, or any other conditions that indicate a deficiency that could affect the integrity of the portable tank. The extent of the exceptional inspection and test shall depend on the amount of damage or deterioration of the portable tank. It shall include at least the 2.5-year periodic inspection and test according to 6.7.4.14.4.
- 6.7.4.14.8** The internal examination during the initial inspection and test shall ensure that the shell is inspected for pitting, corrosion, or abrasions, dents, distortions, defects in welds or any other conditions that might render the portable tank unsafe for transport.
- 6.7.4.14.9** The external examination shall ensure that:
- .1 the external piping, valves, pressurizing/cooling systems when applicable, and gaskets are inspected for corroded areas, defects, or any other conditions, including leakage, that might render the portable tank unsafe for filling, discharge or transport;
  - .2 there is no leakage at any manhole covers or gaskets;
  - .3 missing or loose bolts or nuts on any flanged connection or blank flange are replaced or tightened;
  - .4 all emergency devices and valves are free from corrosion, distortion and any damage or defect that could prevent their normal operation. Remote closure devices and self-closing stop-valves shall be operated to demonstrate proper operation;
  - .5 required marks on the portable tank are legible and in accordance with the applicable provisions; and
  - .6 the framework, the supports and the arrangements for lifting the portable tank are in satisfactory condition.
- 6.7.4.14.10** The inspections and tests in 6.7.4.14.1, 6.7.4.14.3, 6.7.4.14.4 and 6.7.4.14.7 shall be performed or witnessed by an expert approved by the competent authority or its authorized body. When the pressure test is a part of the inspection and test, the test pressure shall be the one indicated on the data plate of the portable tank. While under pressure, the portable tank shall be inspected for any leaks in the shell, piping or equipment.
- 6.7.4.14.11** In all cases when cutting, burning or welding operations on the shell of a portable tank have been effected, that work shall be to the approval of the competent authority or its authorized body, taking into account the pressure-vessel code used for the construction of the shell. A pressure test to the original test pressure shall be performed after the work is completed.
- 6.7.4.14.12** When evidence of any unsafe condition is discovered, the portable tank shall not be returned to service until it has been corrected and the test is repeated and passed.

#### 6.7.4.15 Marking

- 6.7.4.15.1** Every portable tank shall be fitted with a corrosion-resistant metal plate permanently attached to the portable tank in a conspicuous place readily accessible for inspection. When for reasons of portable tank arrangements the plate cannot be permanently attached to the shell, the shell shall be marked with at least the information required by the pressure-vessel code. As a minimum, at least the following information shall be marked on the plate by stamping or by any other similar method:
- (a) owner information
    - (i) owner's registration number;
  - (b) manufacturing information
    - (i) country of manufacture;
    - (ii) year of manufacture;
    - (iii) manufacturer's name or mark;
    - (iv) manufacturer's serial number;

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## (c) approval information

## (i) the United Nations packaging symbol:



This symbol shall not be used for any purpose other than certifying that a packaging, a flexible bulk container, a portable tank or a MEGC complies with the relevant requirements in chapters 6.1, 6.2, 6.3, 6.5, 6.6, 6.7 or 6.9;

## (ii) approval country;

## (iii) authorized body for the design approval;

## (iv) design approval number;

## (v) letters "AA", if the design was approved under alternative arrangements (see 6.7.1.2);

## (vi) pressure-vessel code to which the shell is designed;

## (d) pressures

## (i) MAWP (in bar gauge or kPa gauge);\*

## (ii) test pressure (in bar gauge or kPa gauge);\*

## (iii) initial pressure test date (month and year);

## (iv) identification mark of the initial pressure test witness;

## (e) temperatures

## (i) minimum design temperature (in °C);\*

## (f) materials

## (i) shell material(s) and material standard reference(s);

## (ii) equivalent thickness in reference steel (in mm);\*

## (g) capacity

## (i) tank water capacity at 20°C (in litres);\*

## (h) insulation

## (i) either "Thermally insulated" or "Vacuum insulated" (as applicable);

## (ii) effectiveness of the insulation system (heat influx) (in Watts);\*

## (i) holding times – for each refrigerated liquefied gas permitted to be transported in the portable tank:

## (i) name, in full, of the refrigerated liquefied gas;

## (ii) reference holding time (in days or hours);\*

## (iii) initial pressure (in bar gauge or kPa gauge);\*

## △ (iv) maximum allowable mass of gas filled (in kg);\*

## (j) periodic inspections and tests

## (i) type of the most recent periodic test (2.5-year, 5-year or exceptional);

## (ii) date of the most recent periodic test (month and year);

## (iii) identification mark of the authorized body who performed or witnessed the most recent test.

\* The unit used shall be indicated.

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Figure 6.7.4.15.1 – Example of a plate for marking

|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
|-----------------------------------------------------------------------------------|------------------------------------------|---------------|------------------------|----------------|------------------|----------------------|--------------------------------------|
| Owner's registration number                                                       |                                          |               |                        |                |                  |                      |                                      |
| <b>MANUFACTURING INFORMATION</b>                                                  |                                          |               |                        |                |                  |                      |                                      |
| Country of manufacture                                                            |                                          |               |                        |                |                  |                      |                                      |
| Year of manufacture                                                               |                                          |               |                        |                |                  |                      |                                      |
| Manufacturer                                                                      |                                          |               |                        |                |                  |                      |                                      |
| Manufacturer's serial number                                                      |                                          |               |                        |                |                  |                      |                                      |
| <b>APPROVAL INFORMATION</b>                                                       |                                          |               |                        |                |                  |                      |                                      |
|  | Approval country                         |               |                        |                |                  |                      |                                      |
|                                                                                   | Authorized body for design approval      |               |                        |                |                  |                      |                                      |
|                                                                                   | Design approval number                   |               |                        |                |                  | "AA" (if applicable) |                                      |
| Shell design code (pressure-vessel code)                                          |                                          |               |                        |                |                  |                      |                                      |
| <b>PRESSURES</b>                                                                  |                                          |               |                        |                |                  |                      |                                      |
| MAWP                                                                              |                                          |               |                        | bar or kPa     |                  |                      |                                      |
| Test pressure                                                                     |                                          |               |                        | bar or kPa     |                  |                      |                                      |
| Initial pressure test date:                                                       |                                          | (mm/yyyy)     |                        | Witness stamp: |                  |                      |                                      |
| <b>TEMPERATURES</b>                                                               |                                          |               |                        |                |                  |                      |                                      |
| Minimum design temperature                                                        |                                          |               |                        | °C             |                  |                      |                                      |
| <b>MATERIALS</b>                                                                  |                                          |               |                        |                |                  |                      |                                      |
| Shell material(s) and material standard reference(s)                              |                                          |               |                        |                |                  |                      |                                      |
| Equivalent thickness in reference steel                                           |                                          |               |                        | mm             |                  |                      |                                      |
| <b>CAPACITY</b>                                                                   |                                          |               |                        |                |                  |                      |                                      |
| Tank water capacity at 20°C                                                       |                                          |               |                        | litres         |                  |                      |                                      |
| <b>INSULATION</b>                                                                 |                                          |               |                        |                |                  |                      |                                      |
| "Thermally insulated" or "Vacuum insulated" (as applicable)                       |                                          |               |                        |                |                  |                      |                                      |
| Heat influx                                                                       |                                          |               |                        | Watts          |                  |                      |                                      |
| <b>HOLDING TIMES</b>                                                              |                                          |               |                        |                |                  |                      |                                      |
| △                                                                                 | Refrigerated liquefied gas(es) permitted |               | Reference holding time |                | Initial pressure |                      | Maximum allowable mass of gas filled |
|                                                                                   |                                          |               | days or hours          |                | bar or kPa       |                      | kg                                   |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
| <b>PERIODIC INSPECTIONS/TESTS</b>                                                 |                                          |               |                        |                |                  |                      |                                      |
| Test type                                                                         | Test date                                | Witness stamp | Test type              | Test date      | Witness stamp    |                      |                                      |
|                                                                                   | (mm/yyyy)                                |               |                        | (mm/yyyy)      |                  |                      |                                      |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |
|                                                                                   |                                          |               |                        |                |                  |                      |                                      |



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- 6.7.4.15.2** The following information shall be durably marked either on the portable tank itself or on a metal plate firmly secured to the portable tank:
- Name of the owner and the operator
  - Name of the refrigerated liquefied gas being transported (and minimum mean bulk temperature)
  - Maximum permissible gross mass (MPGM) . . . . . kg
  - Unladen (tare) mass . . . . . kg
  - Actual holding time for gas being transported . . . . . days (or hours)
  - Portable tank instruction in accordance with 4.2.5.2.6.

- 6.7.4.15.3** If a portable tank is designed and approved for handling in open seas, the words “OFFSHORE PORTABLE TANK” shall be marked on the identification plate.

## **6.7.5 Provisions for the design, construction, inspection and testing of multiple-element gas containers (MEGCs) intended for the transport of non-refrigerated gases**

### **6.7.5.1 Definitions**

For the purposes of this section:

*Elements* are cylinders, tubes or bundles of cylinders;

*Leakproofness test* means a test, using gas, subjecting the elements and the service equipment of the MEGC to an effective internal pressure of not less than 20% of the test pressure;

*Manifold* means an assembly of piping and valves connecting the filling and/or discharge openings of the elements;

*Maximum permissible gross mass (MPGM)* means the sum of the tare mass of the MEGC and the heaviest load authorized for transport;

*Service equipment* means measuring instruments and filling, discharge, venting and safety devices;

*Structural equipment* means the reinforcing, fastening, protective and stabilizing members external to the elements.

### **6.7.5.2 General design and construction provisions**

- 6.7.5.2.1** The MEGC shall be capable of being filled and discharged without the removal of its structural equipment. It shall possess stabilizing members external to the elements to provide structural integrity for handling and transport. MEGCs shall be designed and constructed with supports to provide a secure base during transport and with lifting and tie-down attachments which are adequate for lifting the MEGC, including when loaded to its maximum permissible gross mass. The MEGC shall be designed to be loaded onto or into a vehicle or ship and shall be equipped with skids, mountings or accessories to facilitate mechanical handling.

- 6.7.5.2.2** MEGCs shall be designed, manufactured and equipped in such a way as to withstand all conditions to which they will be subjected during normal conditions of handling and transport. The design shall take into account the effects of dynamic loading and fatigue.

- 6.7.5.2.3** Elements of an MEGC shall be made of seamless steel or composite construction and be constructed and tested according to chapter 6.2. All of the elements in an MEGC shall be of the same design type.

- 6.7.5.2.4** Elements of MEGCs, fittings and pipework shall be:

- △ .1 compatible with the substances intended to be transported (for gases, see ISO 11114-1:2020 and ISO 11114-2:2021); or
- .2 properly passivated or neutralized by chemical reaction.

- 6.7.5.2.5** Contact between dissimilar metals which could result in damage by galvanic action shall be avoided.

- 6.7.5.2.6** The materials of the MEGC, including any devices, gaskets, and accessories, shall not adversely affect the gases intended for transport in the MEGC.

- 6.7.5.2.7** MEGCs shall be designed to withstand, without loss of contents, at least the internal pressure due to the contents, and the static, dynamic and thermal loads during normal conditions of handling and transport. The design shall demonstrate that the effects of fatigue, caused by repeated application of these loads through the expected life of the multiple-element gas container, have been taken into account.

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- 6.7.5.2.8** MEGCs and their fastenings shall, under the maximum permissible load, be capable of withstanding the following separately applied static forces:
- .1 in the direction of travel: twice the MPGM multiplied by the acceleration due to gravity (g);\*
  - .2 horizontally at right angles to the direction of travel: the MPGM (when the direction of travel is not clearly determined, the forces shall be equal to twice the MPGM) multiplied by the acceleration due to gravity (g);\*
  - .3 vertically upwards: the MPGM multiplied by the acceleration due to gravity (g);\* and
  - .4 vertically downwards: twice the MPGM (total loading including the effect of gravity) multiplied by the acceleration due to gravity (g).\*
- 6.7.5.2.9** Under the forces defined above, the stress at the most severely stressed point of the elements shall not exceed the values given in either the relevant standards of 6.2.2.1 or, if the elements are not designed, constructed and tested according to those standards, in the technical code or standard recognized or approved by the competent authority of the country of use (see 6.2.3.1).
- 6.7.5.2.10** Under each of the forces in 6.7.5.2.8, the safety factor for the framework and fastenings to be observed shall be as follows:
- .1 for steels having a clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed yield strength; or
  - .2 for steels with no clearly defined yield point, a safety factor of 1.5 in relation to the guaranteed 0.2% proof strength and, for austenitic steels, the 1% proof strength.
- 6.7.5.2.11** MEGCs intended for the transport of flammable gases shall be capable of being electrically earthed.
- 6.7.5.2.12** The elements shall be secured in a manner that prevents undesired movement in relation to the structure and the concentration of harmful localized stresses.
- 6.7.5.3 Service equipment**
- 6.7.5.3.1** Service equipment shall be configured or designed to prevent damage that could result in the release of the pressure receptacle contents during normal conditions of handling and transport. When the connection between the frame and the elements allows relative movement between the sub-assemblies, the equipment shall be so fastened as to permit such movement without damage to working parts. The manifolds, the discharge fittings (pipe sockets, shut-off devices), and the stop-valves shall be protected from being wrenched off by external forces. Manifold piping leading to shut-off valves shall be sufficiently flexible to protect the valves and the piping from shearing, or releasing the pressure receptacle contents. The filling and discharge devices (including flanges or threaded plugs) and any protective caps shall be capable of being secured against unintended opening.
- 6.7.5.3.2** Each element intended for the transport of gases of class 2.3 shall be fitted with a valve. The manifold for liquefied gases of class 2.3 shall be so designed that the elements can be filled separately and be kept isolated by a valve capable of being sealed. For the transport of gases of class 2.1, the elements shall be divided into groups of not more than 3,000 L each isolated by a valve.
- 6.7.5.3.3** For filling and discharge openings of the MEGC, two valves in series shall be placed in an accessible position on each discharge and filling pipe. One of the valves may be a non-return valve. The filling and discharge devices may be fitted to a manifold. For sections of piping which can be closed at both ends and where a liquid product can be trapped, a pressure relief valve shall be provided to prevent excessive pressure build-up. The main isolation valves on an MEGC shall be clearly marked to indicate their directions of closure. Each stop-valve or other means of closure shall be designed and constructed to withstand a pressure equal to or greater than 1.5 times the test pressure of the MEGC. All stop-valves with screwed spindles shall close by a clockwise motion of the handwheel. For other stop-valves, the positions (open and closed) and direction of closure shall be clearly indicated. All stop-valves shall be designed and positioned to prevent unintentional opening. Ductile metals shall be used in the construction of valves or accessories.
- 6.7.5.3.4** Piping shall be designed, constructed and installed so as to avoid damage due to expansion and contraction, mechanical shock and vibration. Joints in tubing shall be brazed or have an equally strong metal union. The melting point of brazing materials shall be no lower than 525°C. The rated pressure of the service equipment and of the manifold shall be not less than two thirds of the test pressure of the elements.
- 6.7.5.4 Pressure relief devices**
- 6.7.5.4.1** The elements of MEGCs used for the transport of UN 1013 carbon dioxide and UN 1070 nitrous oxide shall be divided into groups of not more than 3,000 L each isolated by a valve. Each group shall be fitted with one

\* For calculation purposes,  $g = 9.81 \text{ m/s}^2$ .

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or more pressure relief devices. If so required by the competent authority of the country of use, MEGCs for other gases shall be fitted with pressure relief devices as specified by that competent authority.

**6.7.5.4.2** When pressure relief devices are fitted, every element or group of elements of an MEGC that can be isolated shall then be fitted with one or more pressure relief devices. Pressure relief devices shall be of a type that will resist dynamic forces, including liquid surge, and shall be designed to prevent the entry of foreign matter, the leakage of gas and the development of any dangerous excess pressure.

**6.7.5.4.3** MEGCs used for the transport of certain non-refrigerated gases identified in instruction T50 in 4.2.5.2.6 may have a pressure relief device as required by the competent authority of the country of use. Unless an MEGC in dedicated service is fitted with an approved pressure relief device constructed of materials compatible with the load, such a device shall comprise a frangible disc preceding a spring-loaded device. The space between the frangible disc and the spring-loaded device may be equipped with a pressure gauge or a suitable tell-tale indicator. This arrangement permits the detection of disc rupture, pinholing or leakage which could cause a malfunction of the pressure relief device. The frangible disc shall rupture at a nominal pressure 10% above the start-to-discharge pressure of the spring-loaded device.

**6.7.5.4.4** In the case of multi-purpose MEGCs used for the transport of low-pressure liquefied gases, the pressure relief devices shall open at a pressure as specified in 6.7.3.7.1 for the gas having the highest maximum allowable working pressure of the gases allowed to be transported in the MEGC.

### **6.7.5.5 Capacity of pressure relief devices**

**6.7.5.5.1** The combined delivery capacity of the pressure relief devices when fitted shall be sufficient that, in the event of complete fire engulfment of the MEGC, the pressure (including accumulation) inside the elements does not exceed 120% of the set pressure of the pressure relief device. The formula provided in CGA S-1.2-2003 *Pressure Relief Device Standards, Part 2, Cargo and Portable Tanks for Compressed Gases* shall be used to determine the minimum total flow capacity for the system of pressure relief devices. CGA S-1.1-2003 *Pressure Relief Device Standards, Part 1, Cylinders for Compressed Gases* may be used to determine the relief capacity of individual elements. Spring-loaded pressure relief devices may be used to achieve the full relief capacity prescribed in the case of low-pressure liquefied gases. In the case of multi-purpose MEGCs, the combined delivery capacity of the pressure relief devices shall be taken for the gas which requires the highest delivery capacity of the gases allowed to be transported in the MEGC.

**6.7.5.5.2** To determine the total required capacity of the pressure relief devices installed on the elements for the transport of liquefied gases, the thermodynamic properties of the gas shall be considered (see, for example, CGA S-1.2-2003 *Pressure Relief Device Standards, Part 2, Cargo and Portable Tanks for Compressed Gases* for low-pressure liquefied gases and CGA S-1.1-2003 *Pressure Relief Device Standards, Part 1, Cylinders for Compressed Gases* for high-pressure liquefied gases).

### **6.7.5.6 Marking of pressure relief devices**

**6.7.5.6.1** Pressure relief devices shall be clearly and permanently marked with the following:

- (a) the manufacturer's name and relevant catalogue number;
- (b) the set pressure and/or the set temperature;
- (c) the date of the last test; and
- (d) the cross sectional flow areas of the spring-loaded pressure relief devices and frangible discs in mm<sup>2</sup>.

**6.7.5.6.2** The rated flow capacity marked on spring-loaded pressure relief devices for low-pressure liquefied gases shall be determined according to ISO 4126-1:2004 and ISO 4126-7:2004.

### **6.7.5.7 Connections to pressure relief devices**

**6.7.5.7.1** Connections to pressure relief devices shall be of sufficient size to enable the required discharge to pass unrestricted to the pressure relief device. No stop-valve shall be installed between the element and the pressure relief devices, except when duplicate devices are provided for maintenance or other reasons, and the stop-valves serving the devices actually in use are locked open, or the stop-valves are interlocked so that at least one of the duplicate devices is always operable and capable of meeting the requirements of 6.7.5.5. There shall be no obstruction in an opening leading to or leaving from a vent or pressure relief device which might restrict or cut off the flow from the element to that device. The opening through all piping and fittings shall have at least the same flow area as the inlet of the pressure relief device to which it is connected. The nominal size of the discharge piping shall be at least as large as that of the pressure relief device outlet. Vents from the pressure relief devices, when used, shall deliver the relieved vapour or liquid to the atmosphere in conditions of minimum back-pressure on the relieving device.

**6.7.5.8 Siting of pressure relief devices**

**6.7.5.8.1** Each pressure relief device shall, under maximum filling conditions, be in communication with the vapour space of the elements for the transport of liquefied gases. The devices, when fitted, shall be so arranged as to ensure that the escaping vapour is discharged upwards and unrestrictedly so as to prevent any impingement of escaping gas or liquid upon the MEGC, its elements or personnel. For flammable, pyrophoric and oxidizing gases, the escaping gas shall be directed away from the element in such a manner that it cannot impinge upon the other elements. Heat-resistant protective devices which deflect the flow of gas are permissible provided the required pressure relief device capacity is not reduced.

**6.7.5.8.2** Arrangements shall be made to prevent access to the pressure relief devices by unauthorized persons and to protect the devices from damage caused by the MEGC overturning.

**6.7.5.9 Gauging devices**

**6.7.5.9.1** When an MEGC is intended to be filled by mass, it shall be equipped with one or more gauging devices. Level-gauges made of glass or other fragile material shall not be used.

**6.7.5.10 MEGC supports, frameworks, lifting and tie-down attachments**

**6.7.5.10.1** MEGCs shall be designed and constructed with a support structure to provide a secure base during transport. The forces specified in 6.7.5.2.8 and the safety factor specified in 6.7.5.2.10 shall be considered in this aspect of the design. Skids, frameworks, cradles or other similar structures are acceptable.

**6.7.5.10.2** The combined stresses caused by element mountings (e.g. cradles, frameworks, etc.) and MEGC lifting and tie-down attachments shall not cause excessive stress in any element. Permanent lifting and tie-down attachments shall be fitted to all MEGCs. In no case shall mountings or attachments be welded onto the elements.

**6.7.5.10.3** In the design of supports and frameworks, the effects of environmental corrosion shall be taken into account.

**6.7.5.10.4** When MEGCs are not protected during transport, according to 4.2.4.3, the elements and service equipment shall be protected against damage resulting from lateral or longitudinal impact or overturning. External fittings shall be protected so as to preclude the release of the elements' contents upon impact or overturning of the MEGC on its fittings. Particular attention shall be paid to the protection of the manifold. Examples of protection include:

- .1 protection against lateral impact, which may consist of longitudinal bars;
- .2 protection against overturning, which may consist of reinforcement rings or bars fixed across the frame;
- .3 protection against rear impact, which may consist of a bumper or frame;
- .4 protection of the elements and service equipment against damage from impact or overturning by use of an ISO frame in accordance with the relevant provisions of ISO 1496-3:1995.

**6.7.5.11 Design approval**

**6.7.5.11.1** The competent authority or its authorized body shall issue a design approval certificate for any new design of an MEGC. This certificate shall attest that the MEGC has been surveyed by that authority, is suitable for its intended purpose and meets the requirements of this chapter, the applicable provisions for gases of chapter 4.1 and of packing instruction P200. When a series of MEGCs are manufactured without change in the design, the certificate shall be valid for the entire series. The certificate shall refer to the prototype test report, the materials of construction of the manifold, the standards to which the elements are made and an approval number. The approval number shall consist of the distinguishing sign or mark of the country granting the approval, indicated by the distinguishing sign used on vehicles in international road traffic\* and a registration number. Any alternative arrangements according to 6.7.1.2 shall be indicated on the certificate. A design approval may serve for the approval of smaller MEGCs made of materials of the same type and thickness, by the same fabrication techniques and with identical supports, equivalent closures and other appurtenances.

**6.7.5.11.2** The prototype test report for the design approval shall include at least the following:

- .1 the results of the applicable framework test specified in ISO 1496-3:1995;
- .2 the results of the initial inspection and test specified in 6.7.5.12.3;
- .3 the results of the impact test specified in 6.7.5.12.1; and
- .4 certification documents verifying that the cylinders and tubes comply with the applicable standards.

\* Distinguishing sign of the State of registration used on motor vehicles and trailers in international road traffic, e.g. in accordance with the Geneva Convention on Road Traffic of 1949 or the Vienna Convention on Road Traffic of 1968.

**Part 6 – Construction and testing of packagings, IBCs, etc.****6.7.5.12 Inspection and testing**

- 6.7.5.12.1** MEGCs meeting the definition of *container* in the *International Convention for Safe Containers, 1972* (CSC Convention), as amended, shall not be used unless they are successfully qualified by subjecting a representative prototype of each design to the Dynamic, Longitudinal Impact Test prescribed in the *Manual of Tests and Criteria*, part IV, section 41. This provision only applies to MEGCs which are constructed according to a design approval certificate which has been issued on or after 1 January 2008.
- 6.7.5.12.2** The elements and items of equipment of each MEGC shall be inspected and tested before being put into service for the first time (initial inspection and test). Thereafter, MEGCs shall be inspected at no more than five-year intervals (5-year periodic inspection). An exceptional inspection and test shall be performed, regardless of the last periodic inspection and test, when necessary according to 6.7.5.12.5.
- 6.7.5.12.3** The initial inspection and test of an MEGC shall include a check of the design characteristics, an external examination of the MEGC and its fittings with due regard to the gases to be transported, and a pressure test performed at the test pressures according to packing instruction P200. The pressure test of the manifold may be performed as a hydraulic test or by using another liquid or gas with the agreement of the competent authority or its authorized body. Before the MEGC is placed into service, a leakproofness test and a test of the satisfactory operation of all service equipment shall also be performed. When the elements and their fittings have been pressure-tested separately, they shall be subjected together after assembly to a leakproofness test.
- 6.7.5.12.4** The 5-year periodic inspection and test shall include an external examination of the structure, the elements and the service equipment in accordance with 6.7.5.12.6. The elements and the piping shall be tested at the periodicity specified in packing instruction P200 and in accordance with the provisions described in 6.2.1.6. When the elements and equipment have been pressure-tested separately, they shall be subjected together after assembly to a leakproofness test.
- 6.7.5.12.5** An exceptional inspection and test is necessary when the MEGC shows evidence of damaged or corroded areas, leakage, or other conditions that indicate a deficiency that could affect the integrity of the MEGC. The extent of the exceptional inspection and test shall depend on the amount of damage or deterioration of the MEGC. It shall include at least the examinations required under 6.7.5.12.6.
- 6.7.5.12.6** The examinations shall ensure that:
- .1 the elements are inspected externally for pitting, corrosion, abrasions, dents, distortions, defects in welds or any other conditions, including leakage, that might render the MEGC unsafe for transport;
  - .2 the piping, valves, and gaskets are inspected for corroded areas, defects, and other conditions, including leakage, that might render the MEGC unsafe for filling, discharge or transport;
  - .3 missing or loose bolts or nuts on any flanged connection or blank flange are replaced or tightened;
  - .4 all emergency devices and valves are free from corrosion, distortion and any damage or defect that could prevent their normal operation. Remote closure devices and self-closing stop-valves shall be operated to demonstrate proper operation;
  - .5 required marks on the MEGC are legible and in accordance with the applicable requirements; and
  - .6 the framework, the supports and the arrangements for lifting the MEGC are in satisfactory condition.
- 6.7.5.12.7** The inspections and tests in 6.7.5.12.1, 6.7.5.12.3, 6.7.5.12.4 and 6.7.5.12.5 shall be performed or witnessed by a body authorized by the competent authority. When the pressure test is a part of the inspection and test, the test pressure shall be the one indicated on the data plate of the MEGC. While under pressure, the MEGC shall be inspected for any leaks in the elements, piping or equipment.
- 6.7.5.12.8** When evidence of any unsafe condition is discovered, the MEGC shall not be returned to service until it has been corrected and the applicable tests and verifications are passed.

**6.7.5.13 Marking**

- 6.7.5.13.1** Every MEGC shall be fitted with a corrosion-resistant metal plate permanently attached to the MEGC in a conspicuous place readily accessible for inspection. The metal plate shall not be affixed to the elements. The elements shall be marked in accordance with chapter 6.2. As a minimum, at least the following information shall be marked on the plate by stamping or by any other similar method:
- (a) owner information
    - (i) owner's registration number;
  - (b) manufacturing information
    - (i) country of manufacture;
    - (ii) year of manufacture;
    - (iii) manufacturer's name or mark;

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- (iv) manufacturer's serial number;
- (c) approval information
  - (i) the United Nations packaging symbol:



This symbol shall not be used for any purpose other than certifying that a packaging, a flexible bulk container, a portable tank or a MEGC complies with the relevant requirements in chapters 6.1, 6.2, 6.3, 6.5, 6.6, 6.7 or 6.9;

- (ii) approval country;
  - (iii) authorized body for the design approval;
  - (iv) design approval number;
  - (v) letters "AA", if the design was approved under alternative arrangements (see 6.7.1.2);
- (d) pressures
  - (i) test pressure (in bar gauge);\*
  - (ii) initial pressure test date (month and year);
  - (iii) identification mark of the initial pressure test witness;
- (e) temperatures
  - (i) design temperature range (in °C);\*
- (f) elements/capacity
  - (i) number of elements;
  - (ii) total water capacity (in litres);\*
- (g) periodic inspections and tests
  - (i) type of the most recent periodic test (5-year or exceptional);
  - (ii) date of the most recent periodic test (month and year);
  - (iii) identification mark of the authorized body who performed or witnessed the most recent test.

\* The unit used shall be indicated.



Figure 6.7.5.13.1 – Example of a plate for marking

|                                                                                   |                                     |                |                      |
|-----------------------------------------------------------------------------------|-------------------------------------|----------------|----------------------|
| Owner's registration number                                                       |                                     |                |                      |
| <b>MANUFACTURING INFORMATION</b>                                                  |                                     |                |                      |
| Country of manufacture                                                            |                                     |                |                      |
| Year of manufacture                                                               |                                     |                |                      |
| Manufacturer                                                                      |                                     |                |                      |
| Manufacturer's serial number                                                      |                                     |                |                      |
| <b>APPROVAL INFORMATION</b>                                                       |                                     |                |                      |
|  | Approval country                    |                |                      |
|                                                                                   | Authorized body for design approval |                |                      |
|                                                                                   | Design approval number              |                | "AA" (if applicable) |
| <b>PRESSURES</b>                                                                  |                                     |                |                      |
| Test pressure                                                                     |                                     | bar            |                      |
| Initial pressure test date:                                                       | (mm/yyyy)                           | Witness stamp: |                      |
| <b>TEMPERATURES</b>                                                               |                                     |                |                      |
| Design temperature range                                                          |                                     | °C to °C       |                      |
| <b>ELEMENTS/CAPACITY</b>                                                          |                                     |                |                      |
| Number of elements                                                                |                                     |                |                      |
| Total water capacity                                                              |                                     | litres         |                      |
| <b>PERIODIC INSPECTIONS/TESTS</b>                                                 |                                     |                |                      |
| Test type                                                                         | Test date                           | Witness stamp  | Test type            |
|                                                                                   | (mm/yyyy)                           |                | (mm/yyyy)            |
|                                                                                   |                                     |                |                      |
|                                                                                   |                                     |                |                      |
|                                                                                   |                                     |                |                      |
|                                                                                   |                                     |                |                      |

## 6.7.5.13.2 The following information shall be durably marked on a metal plate firmly secured to the MEGC:

Name of the operator

Maximum permissible load mass . . . . . kg

Working pressure at 15°C: . . . . . bar gauge

Maximum permissible gross mass (MPGM) . . . . . kg

Unladen (tare) mass . . . . . kg